Measuring the Financial Cycle: A Dynamic Factor Model Approach*

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Abstract

This paper proposes a new country-specific indicator of the financial cycle constructed using a dynamic factor model (DFM). The indicator captures the common co-movement in credit, asset prices, indebtedness, and financial conditions, and can be decomposed into contributions from individual variables to identify the main drivers of risk. We propose using the asymptotic distribution of the principal components estimator to identify periods consistent with a neutral risk environment, which we define as periods where underlying variables are close to their long-run averages. Early-warning exercises suggest that the proposed indicator performs well. In a simulated out-of-sample exercise, the predicted probability of a financial crisis roughly doubles in the year preceding the Global Financial Crisis. These results support its use as a transparent and operational tool for monitoring cyclical systemic risk and informing macroprudential policy.

JEL codes: C54; E44; E58; G01.

Keywords: Financial cycle; Dynamic factor model; Cyclical systemic risk.

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1 Introduction

The 2008 Global Financial Crisis (GFC) underscored the link between the financial sector and the broader economy, placing the concept of the *financial cycle* at the forefront of academic and policy discussions. The financial cycle captures the self-reinforcing interactions between perceptions of value, risk-taking behaviour, and financing constraints, typically manifesting as periods of credit expansion followed by busts (Borio, 2014). The expansion of the financial cycle, often reflected in rapid growth in credit, asset prices, and leverage, can lead to the accumulation of cyclical systemic risk and pose a threat to financial stability.

Accurately measuring the financial cycle is therefore central to macroprudential policy, particularly for the calibration of the countercyclical capital buffer (CCyB). The CCyB is designed to ensure that bank capital requirements reflect the prevailing level of cyclical systemic risk. During the expansionary phase of the financial cycle, when credit growth accelerates and asset prices increase, a gradual build-up of capital strengthens the resilience of the financial system. Conversely, during downturns, releasing the CCyB supports the flow of credit by allowing banks to absorb losses while sustaining lending, thereby mitigating the amplification of adverse shocks to the real economy. In practice, however, cyclical systemic risk is difficult to assess in real time, and macroprudential policy tools such as the CCyB are subject to implementation lags. For this reason, many jurisdictions have adopted a positive neutral CCyB rate even during periods of moderate systemic risk, ensuring that releasable capital is available when financial conditions deteriorate.

Against this backdrop, this paper contributes to the literature on financial cycle measurement by developing a simple methodology to construct country-specific indicators of the financial cycle. The proposed approach captures the common dynamics across key variables reflecting credit developments, asset price fluctuations, leverage, and risk-taking behaviour in the financial system. Specifically, we employ a dynamic factor model (DFM) to extract a latent factor summarising the co-movement of financial variables for a sample of European countries. The use of a common factor is grounded in theoretical frameworks that emphasise self-reinforcing mechanisms in the financial sector, including the financial accelerator (Bernanke et al., 1999), collateral constraints (Kiyotaki and Moore, 1997), and correlated risk-taking by financial intermediaries (He and Krishnamurthy, 2013).

The paper makes three main contributions. First, we propose a new country-specific indicator of the financial cycle, constructed using a dynamic factor model, and show that it tracks the evolution of the financial cycle across countries in a manner consistent with prior expectations from the literature and historical experience. The estimated cycle identifies three prominent peaks (in the early 2000s, in the run-up to the Global Financial Crisis, and following the COVID-19 shock) as well as the prolonged trough associated with post-GFC deleveraging and the renewed contraction from 2022 onwards as monetary policy tightened globally. These patterns align closely with known episodes of financial expansion and stress. The indicator can further be decomposed into contributions from individual variables, providing an interpretable account of the main drivers of financial cycle dynamics at each phase. By revealing which components (credit, asset prices, indebtedness, or financial conditions) are the primary sources of cyclical risk at any given point, the decomposition enhances the transparency of the indicator and allows macroprudential authorities to identify the nature and origin of financial imbalances and to communicate their assessments in a structured manner.

Second, the paper proposes a statistical approach to identifying periods of neutral cyclical systemic risk that serves as a useful complementary tool for macroprudential analysis. Drawing on the asymptotic distribution of the principal components estimator, the approach yields time-varying confidence bands that distinguish elevated, neutral, and subdued financial cycle regimes. This statistical benchmark may provide macroprudential authorities with additional information when assessing whether current cyclical conditions are consistent with a risk-neutral environment. This question is of particular relevance for the calibration of the positive neutral CCyB.

Third, and of direct relevance for policy applications, the indicator is evaluated in a pseudo real-time, out-of-sample exercise designed to replicate the information constraints faced by policymakers in practice. The results point to good early-warning properties. The estimated one-year-ahead probability of a systemic banking crisis approximately doubles in the two years preceding the Global Financial Crisis, rising from around 10% in early 2006 to approximately 20% by 2007. In-sample results further confirm that the DFM-based indicator outperforms the Basel gap, the standard regulatory benchmark, in terms of discriminatory power, as measured by the area under the receiver operating characteristic curve. Together, these findings strengthen the case for using the proposed indicator as an operational tool to monitor cyclical systemic risks.

Related literature. This paper contributes to a broad literature on the measurement of the financial cycle, where a range of approaches have been proposed. A widely used benchmark is the credit-to-GDP gap developed by the Bank for International Settlements (BIS) (Drehmann et al., 2012). While effective in providing early-warning signals of financial crises, its reliance on the Hodrick–Prescott (HP) filter introduces biases in trend estimation (Schüler, 2018). In particular, the expansion phase of the financial cycle tends to be excessively incorporated into the estimated trend, leading to prolonged periods with negative values of the credit-to-GDP gap following episodes of systemic risk build-up. These limitations have motivated alternative methodologies. For example, Drehmann et al. (2012) employ turning-point analysis to identify local extrema, while frequency-based filters reduce reliance on assumptions about the characteristics of the financial cycle but introduce arbitrariness in filter selection. Composite indicators, such as Lang et al. (2019), are designed as early-warning tools for systemic banking crises, whereas Schüler et al. (2020) employ spectral methods to capture co-movement across variables without imposing strong parametric assumptions.

Several studies have applied factor models to analyse the financial cycle, including Breitung and Eickmeier (2016), Menden and Proaño (2017), and Miranda-Agrippino and Rey (2020). However, existing contributions predominantly focus on global financial cycle indicators and their relationship with macroeconomic conditions or monetary policy. In contrast, we develop country-specific measures that are directly relevant for macroprudential policy and explicitly link the financial cycle to cyclical systemic risk and systemic banking crises, with a particular emphasis on early-warning properties.

Organisation of the paper. Section 2 presents the data and the rationale for variable selection. Section 3 outlines the dynamic factor model methodology. Section 4 presents the empirical results, and Section 5 concludes.

2 Data

2.1 Variable selection

Definitions. Borio (2014) defines the financial cycle as “self-reinforcing interactions between perceptions of value and risk, attitudes towards risk, and financing constraints, which translate into booms followed by busts." This definition highlights that the financial cycle is characterised by joint fluctuations in credit, asset prices, and leverage that can become disconnected from underlying fundamentals. As financial imbalances accumulate over time, cyclical systemic risk increases, raising the likelihood of severe downturns and banking crises.

Measuring the financial cycle therefore requires selecting variables that capture both the build-up of financial imbalances and their potential materialisation. In this paper, we focus on four key dimensions: credit, asset prices, indebtedness, and financial conditions. These dimensions are chosen based on their established role in the literature as core components of cyclical systemic risk and their relevance for early-warning analysis.

To preserve interpretability and ensure a transparent link between the estimated factor and its underlying drivers, we deliberately restrict the analysis to a parsimonious set of variables within each dimension. This approach facilitates the economic interpretation of the financial cycle indicator while retaining the key information needed to capture its dynamics.

Credit. The central role of credit in the financial cycle is well established in both theoretical and empirical work. Early contributions by Fisher (1933) and Minsky (1977) highlight the role of credit cycles in generating and amplifying macroeconomic instability, while subsequent work shows how credit acts as a propagation mechanism for shocks (Bernanke, 1983; Gertler, 1988). This mechanism is formalised in the financial accelerator framework of Bernanke et al. (1999), which emphasises the endogenous amplification of shocks through credit market conditions. Empirically, credit aggregates have been shown to be among the most reliable predictors of systemic banking crises (Schularick and Taylor, 2012).

To capture these dynamics, we include two measures of credit: credit to non-financial corporations and credit to households (including non-profit institutions). This distinction allows us to account for differences in borrowing behaviour across sectors, which may have distinct implications for the build-up of cyclical systemic risk. Both series are constructed from the total stock of credit to the private non-financial sector, covering all sources of financing including domestic banks, non-bank financial institutions, non-financial corporations, and cross-border lending. As such, the indicator provides a comprehensive measure of credit developments in the economy.

Asset prices. Asset prices play a central role in the financial cycle, both as indicators of financial imbalances and as channels through which shocks are amplified. In particular, increases in asset prices can relax borrowing constraints and support credit expansion, while subsequent declines can trigger adverse feedback loops by reducing collateral values (Kiyotaki and Moore, 1997). Empirically, asset price dynamics, especially in housing markets, are closely associated with the build-up and materialisation of systemic risk (Jordà et al., 2015; Claessens et al., 2012). House prices, in particular, have been shown to be strong predictors of banking crises, while equity prices often peak prior to crisis episodes (Reinhart and Rogoff, 2013).

To capture these dynamics, we include both house price and share price indices. The house price index reflects inflation-adjusted transaction prices of residential property, providing a measure of developments in real estate markets that are closely linked to credit conditions. Share price indices capture the prices of publicly traded equities, aggregated from daily closing values, and provide forward-looking information on market valuations and risk sentiment.

Indebtedness. While credit captures the expansion of financing over the financial cycle, indebtedness reflects the accumulation of debt and the associated vulnerabilities in the economy. High levels of leverage increase the sensitivity of households, firms, and financial intermediaries to adverse shocks, as declines in income or asset prices can impair debt servicing capacity and amplify financial distress (Adrian and Shin, 2010; Brunnermeier and Pedersen, 2009). As a result, elevated indebtedness is a key determinant of the severity of downturns and the materialisation of systemic risk.

To capture these dynamics, we include two complementary indicators: the credit-to-GDP ratio and the debt service ratio (DSR). The credit-to-GDP ratio provides a measure of the stock of debt relative to the size of the economy and is widely used as an indicator of systemic risk, with well-documented early-warning properties for banking crises (Aikman et al., 2015). The DSR captures the share of income devoted to debt servicing, providing a direct measure of repayment burden and financial stress faced by households and non-financial corporations. Together, these variables capture both the level and the sustainability of indebtedness over the financial cycle.

Financial conditions. Financial conditions capture the pricing of risk and the level of risk appetite in financial markets, both of which are central to the financial cycle. We proxy these conditions using the spread between each country's 10-year sovereign bond yield and that of Germany, which serves as a benchmark reflecting low credit risk within the European context. Changes in sovereign spreads provide information on shifts in perceived risk and financing conditions, as increases in risk premia tend to coincide with periods of financial stress.

This measure is closely related to indicators used in financial stress indices (Hubrich and Tetlow, 2015), which distinguish periods of low risk appetite from episodes of heightened uncertainty. Moreover, sovereign bond spreads are informative about broader financial conditions, as movements in sovereign risk premia are often mirrored in borrowing costs for households, firms, and financial intermediaries. Including this variable therefore enhances the ability of the indicator to capture the materialisation of systemic risk, complementing other variables that primarily reflect its build-up.

2.2 Variable treatment

Stationarity. An underlying assumption of dynamic factor models is that the data are covariance stationary. To satisfy this requirement, all variables, except the government bond spread, are transformed into 2-year growth rates, while the bond spread is expressed in 2-year differences. These transformations are guided by unit root tests and case-by-case judgement.

The choice of 2-year growth rates reflects the medium-term nature of the financial cycle and is supported by evidence that longer-horizon transformations exhibit stronger early-warning properties for systemic banking crises (Lang et al., 2019). By focusing on medium-term dynamics, this approach captures the gradual build-up of financial imbalances that precede crisis episodes.

Importantly, this methodology analyses cycles in growth rates rather than relying on gap measures, such as the credit-to-GDP gap, which extract cyclical components from time series levels. In line with Schüler et al. (2020) and Lang et al. (2019), this avoids the spurious cycles and endpoint biases often associated with filtering techniques (Schüler, 2018), providing a more robust representation of financial cycle dynamics.

Low-frequency movements. To isolate the cyclical components of the variables, low-frequency movements are removed from the data. Long-term trends often reflect structural changes and can obscure medium-term fluctuations relevant for the financial cycle. As emphasised by Stock and Watson (2016), removing these low-frequency components is essential for distinguishing persistent trends from cyclical dynamics in macroeconomic time series.

This is particularly relevant in the European context, where greater financial integration and the introduction of the euro have significantly influenced long-term financial dynamics and contributed to the synchronisation of financial cycles across countries (Jordà et al., 2019; Aikman et al., 2015). In addition, the post-GFC period has been marked by significant institutional changes that have reshaped the functioning of financial markets. Accounting for these low-frequency movements is therefore important to avoid conflating structural trends with cyclical fluctuations.

In line with this approach, we apply a filtering method to extract medium-term dynamics (Drehmann et al., 2012; Schüler et al., 2020). Specifically, we use the Butterworth filter proposed by Pollock (2000), which offers greater flexibility than the Hodrick–Prescott (HP) filter in capturing time-varying trends.

We focus on cycles with periodicities between 4 and 25 years. Frequencies longer than 25 years are removed to exclude low-frequency trends, while fluctuations with periodicities shorter than 4 years are filtered out to reduce high-frequency noise. This approach allows us to isolate medium-term financial cycle dynamics in a consistent and robust manner.

Standardisation. All variables are standardised to have zero mean and unit variance. This is a standard step in dynamic factor models, as the estimated factor is sensitive to the scale of the variables. Standardisation ensures comparability across variables and allows the resulting financial cycle indicator to be interpreted in terms of deviations from its historical average. In this framework, large positive or negative values of the indicator reflect significant departures from typical conditions and can be interpreted as signals of financial imbalances.

2.3 Descriptive statistics

Sample. Our sample consists of quarterly data for 10 European countries: Sweden (SE), Italy (IT), France (FR), Germany (DE), Belgium (BE), the Netherlands (NL), Finland (FI), the United Kingdom (GB), Portugal (PT), and Spain (ES). The dataset spans from 1995 Q3 to 2025 Q3. The country selection balances sample length with data availability across countries. Our focus on European countries reflects their exposure to similar economic shocks and similar institutional frameworks, enabling more meaningful comparisons. In the following sections, we construct the financial cycle indicator for each country and present the results by examining the distribution of the indicator across the sample.

Descriptive statistics. Table 1 reports summary statistics for the selected variables, expressed in 2-year growth rates, except for bond spreads, which are presented in 2-year differences. The results reveal substantial heterogeneity across countries and across types of financial indicators.

Credit to non-financial corporations shows moderate average growth across countries, ranging from 7.5% in Italy to 18.4% in the United Kingdom, with relatively limited dispersion. In contrast, credit to households grows more strongly and is more volatile, particularly in the United Kingdom (18.9%), Spain (17.2%), and Portugal (16.9%).

Asset price dynamics display greater cross-country variation. House prices record positive average growth in most countries, with higher values in Spain (8.3%), the Netherlands (7.6%), and the United Kingdom (7.3%), while Italy stands out with a negative average growth rate (−1.1%). Equity prices exhibit robust growth across all countries, reaching 23.6% in Finland and 22.8% in Sweden, but with substantial volatility.

Indicators of indebtedness show more moderate dynamics. Debt service ratios display near-zero average growth across countries, ranging from −1.0% in Portugal to 3.3% in the United Kingdom. By contrast, credit-to-GDP ratios increase more noticeably in countries such as Belgium (4.7%), the United Kingdom (4.2%), Sweden (4.0%), and Spain (3.7%), indicating a gradual accumulation of leverage over the sample period.

Finally, sovereign bond spreads generally decline over the sample period, particularly in Sweden (−28%), Spain (−25%), and Finland (−24%). As Germany serves as the benchmark for the spread measure, it is not directly comparable to other countries in this dimension. Variability in spreads is highest in Portugal, reflecting the pronounced fluctuations observed during the sovereign debt crisis.

Table 1: Summary statistics

	Credit NFC (%)		Credit HH (%)		House prices (%)		Share prices (%)		DSR (%)		Credit-to-GDP (%)		Bond spread	
Country	**Mean**	**SD**	**Mean**	**SD**	**Mean**	**SD**	**Mean**	**SD**	**Mean**	**SD**	**Mean**	**SD**	**Mean**	**SD**
BE	14.3	11.1	12.7	6.1	6.0	5.8	14.2	31.7	3.0	8.4	4.7	7.8	−0.12	0.56
DE	10.6	8.0	9.9	7.6	0.6	6.5	15.3	30.1	0.2	8.5	1.6	4.5	−0.31	1.33
ES	14.9	19.2	17.2	19.6	8.3	18.1	19.1	47.0	−0.3	13.2	3.7	12.6	−0.25	1.36
FI	11.0	11.4	10.7	11.1	0.4	12.1	23.6	65.2	0.9	13.8	3.1	9.2	−0.24	0.71
FR	13.0	7.8	16.2	11.6	3.8	10.0	20.2	34.5	1.1	9.0	3.4	4.8	−0.14	0.93
GB	18.4	17.1	18.9	13.9	7.3	14.1	16.3	22.4	3.3	11.4	4.2	8.6	−0.16	0.83
IT	7.5	8.9	11.5	10.6	−1.1	8.6	14.2	36.1	−0.1	8.4	2.5	8.2	−0.23	1.42
NL	9.9	5.2	13.1	9.8	7.6	11.1	16.0	32.2	−0.7	6.1	2.2	6.1	0.00	0.20
PT	10.4	11.6	16.9	22.0	3.0	9.8	17.5	38.9	−1.0	9.4	3.1	11.2	−0.22	2.63
SE	15.5	14.1	12.8	8.0	6.7	12.2	22.8	34.3	1.7	11.3	4.0	8.4	−0.28	0.85

Notes: NFC refers to non-financial corporations; HH to households. Bond spreads are defined as the difference between the 10-year sovereign bond yield of each country and that of Germany (DE), which serves as the benchmark; for Germany, the reported value corresponds to the 10-year sovereign bond yield itself. All growth-rate variables are expressed as 2-year growth rates; bond spreads are expressed as 2-year differences.

3 Econometric framework

3.1 Representation

Let $\boldsymbol{x}_t = [x_{1,t}, x_{2,t}, \ldots, x_{N,t}]'$, for $t = 1, \ldots, T$, denote a stationary N-dimensional vector process standardised to have mean zero and unit variance. We assume that $\boldsymbol{x}_t$ admits the following factor model representation:

$$\boldsymbol{x}_t = \boldsymbol{\Lambda} \boldsymbol{f}_t + \boldsymbol{e}_t \tag{1}$$

where $\boldsymbol{f}_t$ is an $r \times 1$ vector of unobserved common factors, $\boldsymbol{\Lambda}$ is an $N \times r$ matrix of factor loadings, and $\boldsymbol{e}_t = [e_{1,t}, \ldots, e_{N,t}]'$ denotes the idiosyncratic component. The term $\boldsymbol{\chi}_t = \boldsymbol{\Lambda} \boldsymbol{f}_t$ is referred to as the common component.

The common factors follow a stationary vector autoregressive process of order p:

$$\boldsymbol{f}_t = \sum_{j=1}^{p} \boldsymbol{\Phi}_j \boldsymbol{f}_{t-j} + \boldsymbol{\eta}_t \tag{2}$$

where $\boldsymbol{\eta}_t$ is a vector of mean-zero innovations with covariance matrix $\boldsymbol{\Sigma}_\eta$. The idiosyncratic components $\boldsymbol{e}_t$ are allowed to exhibit weak cross-sectional and serial dependence, giving rise to an approximate factor model. It is assumed that $\boldsymbol{e}_t$ is uncorrelated with $\boldsymbol{\eta}_t$ at all leads and lags.

3.2 Estimation

We consider three approaches to estimate the dynamic factor model: principal components analysis (PCA), quasi-maximum likelihood (QML) estimation, and a two-step estimator. PCA provides a simple and robust benchmark by estimating the factor space from the covariance structure of the data without imposing an explicit dynamic model. QML exploits the state-space representation and incorporates factor dynamics, improving efficiency under stronger assumptions. The two-step estimator combines these approaches by retaining the tractability of PCA while incorporating dynamic structure through state-space methods.

Principal components. PCA is a nonparametric approach, as it does not require specifying a model for the factor dynamics or imposing distributional assumptions on the idiosyncratic components. Under the static representation, the factor loadings $\boldsymbol{\Lambda}$ and the factors $\boldsymbol{f}_t$ can be estimated by minimising:

$$SSR(\boldsymbol{\Lambda}, \boldsymbol{f}_t) = \frac{1}{NT}\sum_{t=1}^{T}(\boldsymbol{x}_t - \boldsymbol{\Lambda}\boldsymbol{f}_t)'(\boldsymbol{x}_t - \boldsymbol{\Lambda}\boldsymbol{f}_t) \tag{3}$$

Due to the indeterminacy of the factors and loadings, identification requires normalisation.[1] A convenient normalisation imposes $N^{-1}\boldsymbol{\Lambda}'\boldsymbol{\Lambda} = \boldsymbol{I}_r$ and assumes that $\mathbb{E}(\boldsymbol{f}_t\boldsymbol{f}_t')$ is diagonal.

Under these conditions, the PCA estimators are obtained from the eigenvalue decomposition of the sample covariance matrix:

$$\hat{\boldsymbol{\Sigma}}_x = \frac{1}{T}\sum_{t=1}^{T}\boldsymbol{x}_t\boldsymbol{x}_t' \tag{4}$$

[1]This reflects the fact that both the factors and loadings are unobserved and only their product is identified. To illustrate the identification issue, consider any invertible $r \times r$ matrix $\boldsymbol{M}$. Then, the common component $\boldsymbol{\Lambda}\boldsymbol{f}_t$ is observationally equivalent to $\boldsymbol{\Lambda}^*\boldsymbol{f}_t^* = (\boldsymbol{\Lambda}\boldsymbol{M}^{-1})(\boldsymbol{M}\boldsymbol{f}_t)$, where $\boldsymbol{\Lambda}^* = \boldsymbol{\Lambda}\boldsymbol{M}^{-1}$ and $\boldsymbol{f}_t^* = \boldsymbol{M}\boldsymbol{f}_t$.

Let $\hat{\boldsymbol{\lambda}}_1, \ldots, \hat{\boldsymbol{\lambda}}_r$ denote the unit-norm eigenvectors of $\hat{\boldsymbol{\Sigma}}_x$ associated with its r largest eigenvalues. The factor loadings estimator $\hat{\boldsymbol{\Lambda}}$ and the estimated factors are then given by:

$$\hat{\boldsymbol{\Lambda}} = \sqrt{N}\,[\hat{\boldsymbol{\lambda}}_1, \ldots, \hat{\boldsymbol{\lambda}}_r], \qquad \hat{\boldsymbol{f}}_t = N^{-1}\hat{\boldsymbol{\Lambda}}'\boldsymbol{x}_t \tag{5}$$

The consistency of the PCA estimator was established by Bai and Ng (2002) and Stock and Watson (2002), while Bai (2003) provides asymptotic distribution theory. After estimating the factors, their dynamics can be characterised by fitting a VAR model. Based on standard information criteria, we select $p = 4$.

Quasi-maximum likelihood (QML) estimation. PCA does not explicitly account for factor dynamics, which may result in efficiency losses in estimation. QML addresses this by exploiting the state-space representation of the model, combining cross-sectional and time-series information.

The model can be written in state-space form with measurement equation:

$$\boldsymbol{x}_t = \boldsymbol{\Lambda}\boldsymbol{f}_t + \boldsymbol{e}_t \tag{6}$$

and transition equation:

$$\boldsymbol{F}_t = \boldsymbol{A}\boldsymbol{F}_{t-1} + \boldsymbol{B}\boldsymbol{\eta}_t \tag{7}$$

where the state vector is defined as:

$$\boldsymbol{F}_t = \begin{bmatrix} \boldsymbol{f}_t \\ \boldsymbol{f}_{t-1} \\ \vdots \\ \boldsymbol{f}_{t-p+1} \end{bmatrix} \tag{8}$$

and the system matrices $\boldsymbol{A}$ and $\boldsymbol{B}$ are given by:

$$\boldsymbol{A} = \begin{bmatrix} \boldsymbol{\Phi}_1 & \cdots & \boldsymbol{\Phi}_p \\ \boldsymbol{I}_r & \cdots & \boldsymbol{0} \\ \vdots & \ddots & \vdots \\ \boldsymbol{0} & \cdots & \boldsymbol{I}_r \end{bmatrix}, \qquad \boldsymbol{B} = \begin{bmatrix} \boldsymbol{I}_r \\ \boldsymbol{0} \\ \vdots \\ \boldsymbol{0} \end{bmatrix} \tag{9}$$

In large cross-sectional settings, full maximum likelihood estimation becomes computationally demanding. Doz et al. (2012) propose a QML approach based on a simplified state-space representation, which assumes that the factors follow a Gaussian distribution and that the idiosyncratic components are mutually orthogonal. Estimation proceeds via the Expectation-Maximisation (EM) algorithm (Watson and Engle, 1983), which alternates between a Kalman filter-based expectation step and a maximisation step based on multivariate regressions.

Importantly, the estimator is consistent as both N and T increase, even when these distributional assumptions are relaxed, including in the presence of non-Gaussian factors and weak cross-sectional dependence in the idiosyncratic components. This robustness motivates its interpretation as a *quasi*-maximum likelihood estimator.

Two-step estimator. A computationally attractive alternative is the two-step estimator of Doz et al. (2011), which avoids the iterative EM optimisation of QML by fixing all model parameters at first-step estimates.

In the first step, PCA yields preliminary loadings $\hat{\boldsymbol{\Lambda}}$, factor estimates $\hat{\boldsymbol{f}}_t$, and residuals from which the diagonal idiosyncratic variance matrix $\hat{\boldsymbol{\Psi}} = \text{diag}(\hat{\sigma}^2_{e,1}, \ldots, \hat{\sigma}^2_{e,N})$ is estimated. A VAR(p) fitted to $\hat{\boldsymbol{f}}_t$ delivers the companion matrix $\hat{\boldsymbol{A}}$ and innovation covariance $\hat{\boldsymbol{\Sigma}}_\eta$.

In the second step, these first-step estimates are treated as fixed parameters and inserted into the state-space representation. The Kalman filter proceeds through a prediction–update recursion, operating on the $rp \times 1$ companion state $\hat{\boldsymbol{F}}_t = [\hat{\boldsymbol{f}}'_t, \ldots, \hat{\boldsymbol{f}}'_{t-p+1}]'$. At each t, the prediction step advances the full state using the estimated VAR:

$$\hat{\boldsymbol{F}}_{t|t-1} = \hat{\boldsymbol{A}}\, \hat{\boldsymbol{F}}_{t-1|t-1} \tag{10}$$

while the update step corrects the current-period factor block $\hat{\boldsymbol{f}}_{t|t-1}$ using the current observation:

$$\hat{\boldsymbol{f}}_{t|t} = \hat{\boldsymbol{f}}_{t|t-1} + \boldsymbol{K}_t \underbrace{\Big(\boldsymbol{x}_t - \hat{\boldsymbol{\Lambda}} \hat{\boldsymbol{f}}_{t|t-1}\Big)}_{\text{prediction error}}, \tag{11}$$

where $\boldsymbol{K}_t$ is the Kalman gain matrix. The prediction error captures the extent to which the current observation departs from what the factor dynamics alone would imply; the gain assigns it the optimal weight relative to the model prediction. A backward smoothing pass then incorporates future observations into each estimate, delivering the full-sample smoothed factor $\hat{\boldsymbol{f}}_{t|T}$.

The result is a factor estimate that draws simultaneously on the cross-sectional co-variation in $\boldsymbol{x}_t$ and the intertemporal structure encoded in the VAR, incorporating the full sample at each t in a single non-iterative pass. Although $\hat{\boldsymbol{\Psi}}$ imposes cross-sectional orthogonality on the idiosyncratic components, Doz et al. (2011) show that the resulting estimates remain consistent, as the bias from this misspecification vanishes asymptotically.

Asymptotic inference. Factor estimates carry estimation uncertainty that must be quantified for inference on the financial cycle. Under the rate condition $\sqrt{N}/T \to 0$, Bai (2003) establishes that, as $(N,T) \to \infty$:

$$\sqrt{N}\left(\hat{\boldsymbol{f}}_t - \boldsymbol{H}'\boldsymbol{f}_t^0\right) \xrightarrow{d} \mathcal{N}\big(\boldsymbol{0},\ \boldsymbol{V}^{-1}\boldsymbol{Q}\boldsymbol{\Gamma}_t\boldsymbol{Q}'\boldsymbol{V}^{-1}\big)\,, \tag{12}$$

where $\boldsymbol{H}$ is the rotation matrix from PCA normalisation, $\boldsymbol{V}$ is the probability limit of the diagonal matrix of the r largest eigenvalues of $(NT)^{-1}\boldsymbol{X}\boldsymbol{X}'$, $\boldsymbol{Q} = \operatorname{plim} \hat{\boldsymbol{F}}'\boldsymbol{F}^0/T$, and $\boldsymbol{\Gamma}_t$ captures the cross-sectional covariance of the idiosyncratic components at time t. A feasible estimator replaces $\boldsymbol{V}$ with its sample analogue $\hat{\boldsymbol{V}}$ and estimates the rotated sandwich $\boldsymbol{Q}\boldsymbol{\Gamma}_t\boldsymbol{Q}'$ as a single unit via $\hat{\boldsymbol{\Gamma}}_t$. The estimator $\hat{\boldsymbol{\Gamma}}_t$ accounts for cross-sectional correlation and heteroskedasticity in the idiosyncratic errors:

$$\hat{\boldsymbol{\Gamma}}_t = \frac{1}{N}\hat{\boldsymbol{\Lambda}}'\hat{\boldsymbol{e}}_t\hat{\boldsymbol{e}}_t'\hat{\boldsymbol{\Lambda}}, \tag{13}$$

where $\hat{\boldsymbol{e}}_t = \boldsymbol{x}_t - \hat{\boldsymbol{\Lambda}}\hat{\boldsymbol{f}}_t$ are the estimated idiosyncratic residuals at time t. The feasible asymptotic variance estimator (Bai, 2003) is then:

$$\widehat{\text{Avar}}(\hat{\boldsymbol{f}}_t) = \hat{\boldsymbol{V}}^{-1}\hat{\boldsymbol{\Gamma}}_t\hat{\boldsymbol{V}}^{-1}, \tag{14}$$

$(1-\alpha)$ confidence bands for the k-th factor are then:

$$\hat{f}_{k,t} \pm z_{\alpha/2}\left[\frac{\widehat{\text{Avar}}(\hat{\boldsymbol{f}}_t)_{kk}}{N}\right]^{1/2}, \tag{15}$$

where $z_{\alpha/2}$ is the upper $\alpha/2$ quantile of the standard normal.

4 Results

4.1 Estimation results

The financial cycle indicator is derived from a dynamic factor model estimated separately for each country, yielding country-specific measures of the financial cycle. In all countries, standard information criteria select a single common factor ($r = 1$), consistent with the objective of capturing the dominant co-movement among the variables. In practice, the estimated factor is closely related to the first principal component of the data, which explains the largest share of their joint variation. To determine the lag order of the factor dynamics, we fit a VAR model to the PCA-estimated factors and select the number of lags based on standard information criteria, which point to $p = 4$ across all countries. We adopt this lag length uniformly across all three estimators (PCA, QML, and the two-step procedure) to ensure comparability of results.

Goodness of fit. Figure 1 reports the proportion of total variance explained by the first factor for each country. The dashed horizontal line represents the cross-country average, showing that the first factor accounts for approximately 44% of the total variance on average. In some countries, this share exceeds 50%, indicating that a single-factor representation provides a reasonably good summary of the common dynamics in the data.

Table 2 reports the R^2 values obtained by regressing each variable on the estimated common component, for all countries in the sample. These values provide a simple measure of how closely each variable co-moves with the financial cycle indicator. The results show that the estimated factor explains a large share of the variation in credit to non-financial corporations, the debt service ratio, and the credit-to-GDP ratio. This finding suggests that the common factor is primarily driven by variables related to credit and indebtedness, which are central to the build-up of cyclical systemic risk.

By contrast, the explanatory power of the factor is generally lower for share prices and sovereign bond spreads, indicating that these variables co-move less systematically with the rest of the dataset. One interpretation is that these series are more strongly

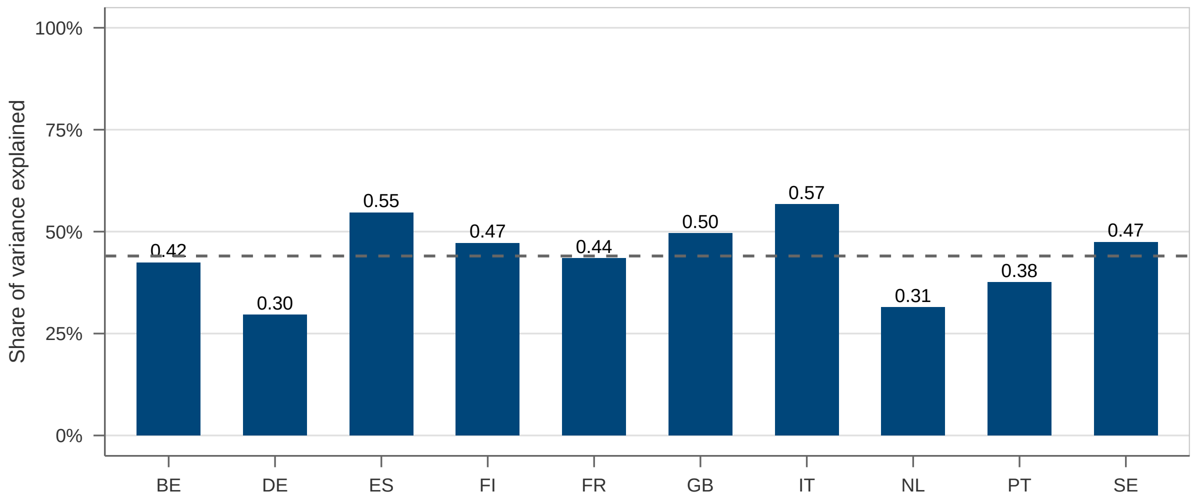

Figure 1: Share of total variance explained by the first factor

Notes: The figure reports the share of total variance explained by the estimated factor of the dynamic factor model estimated separately for each country. The dashed horizontal line indicates the cross-country average. All variables are expressed in 2-year growth rates (2-year differences for bond spreads), filtered to remove low-frequency components, and standardised prior to estimation.

influenced by country-specific shocks, global factors, or abrupt changes in market sentiment, which are not fully captured by a single common factor. This pattern is broadly consistent with Drehmann et al. (2012), who emphasise the central role of credit variables in characterising the financial cycle. At the same time, our results suggest that house prices play a more heterogeneous role across countries and, on average, contribute less to the common factor than credit-based indicators.

Some very low R^2 values also stand out in Table 2, particularly for share prices in Spain, Germany, and the Netherlands, and for house prices in Portugal, Germany, and the Netherlands. These cases likely reflect the fact that, after the transformations and filtering applied to the data, the corresponding series display relatively weak co-movement with the other variables entering the model. In such cases, a single common factor naturally attributes little explanatory power to those series. This does not imply that these variables are unimportant for financial stability in those countries; rather, it indicates that their cyclical dynamics are less aligned with the dominant common component identified by the model.

Evolution of the financial cycle across countries. Figure 2 reports the median financial cycle indicator across the countries in the sample (results for individual countries

Table 2: R^2 of the first factor across variables

Country	Credit NFC	Credit HH	House prices	Share prices	DSR	Credit-to-GDP	Bond spread
BE	0.85	0.15	0.10	0.12	0.84	0.84	0.07
DE	0.66	0.08	0.02	0.00	0.60	0.66	0.06
ES	0.88	0.90	0.48	0.00	0.82	0.75	0.01
FI	0.83	0.31	0.06	0.21	0.89	0.71	0.29
FR	0.83	0.43	0.27	0.01	0.75	0.70	0.05
GB	0.76	0.72	0.39	0.02	0.81	0.76	0.01
IT	0.88	0.80	0.68	0.06	0.70	0.70	0.16
NL	0.62	0.02	0.04	0.00	0.76	0.70	0.05
PT	0.87	0.33	0.00	0.02	0.58	0.72	0.12
SE	0.81	0.66	0.27	0.04	0.75	0.78	0.01
Average	**0.80**	**0.44**	**0.23**	**0.05**	**0.75**	**0.73**	**0.08**

Notes: The table reports the R^2 from regressing each variable on the estimated common factor of the dynamic factor model, computed separately for each country. All variables are expressed in 2-year growth rates (2-year differences for bond spreads), filtered to remove low-frequency components, and standardised prior to estimation. Higher values indicate stronger co-movement with the common factor. HH denotes households.

are provided in Appendix A). Estimates are obtained using three alternative methods: principal components (PCA), quasi-maximum likelihood (QML), and the two-step estimator (TSTEP). The figure also displays the interquartile range (25th–75th percentiles) computed across the full pool of estimates — that is, across all countries and all three estimators at each point in time — capturing both cross-country heterogeneity and variation across estimation methods. The similarity of the median cycles across methods indicates that the main features of the financial cycle are robust to the choice of estimator.

The results show that the financial cycle is characterised by medium-term fluctuations (Borio, 2014; Drehmann et al., 2012), with phases lasting on the order of 8 to 12 years. This distinguishes the financial cycle from the business cycle, which typically operates at higher frequencies (Claessens et al., 2012). Over the sample period, the financial cycle exhibits three prominent peaks: in the early 2000s, in the run-up to the GFC in 2007–2008, and following the COVID-19 shock. These episodes coincide with periods of strong credit expansion, rising leverage, and increasing systemic risk.

Following the GFC, the financial cycle enters a prolonged subdued phase, characterised by lower amplitude and reduced variability. This pattern is consistent with the post-crisis regulatory environment, which includes tighter capital requirements, constraints on leverage, and the broader use of macroprudential policy tools aimed at containing the build-up of systemic risk.

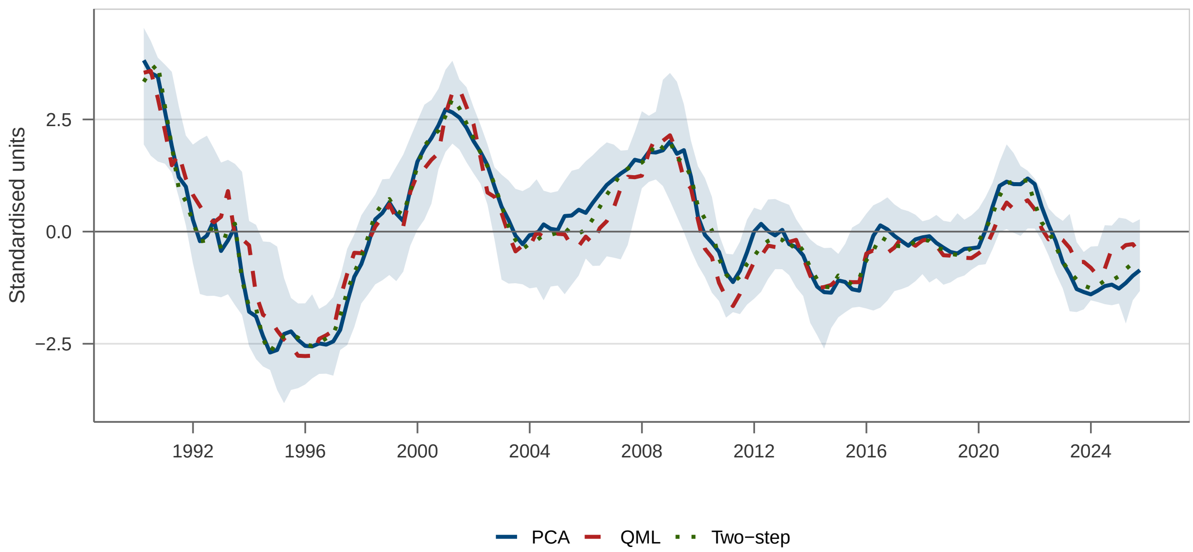

Figure 2: Median financial cycle indicator across countries

Notes: The figure shows the median financial cycle indicator across countries, together with the interquartile range (25th–75th percentiles). Results are reported for three estimation methods: principal components (PCA), quasi-maximum likelihood (QML), and two-step estimator. The shaded area represents this range, computed across the full pool of country-estimator estimates (10 countries × 3 estimators).

The post-COVID period is marked by a sharp but short-lived expansion of the financial cycle, reflecting the unprecedented scale of fiscal and monetary support measures. This phase is associated with rapid credit growth, abundant liquidity, and rising asset prices. However, from 2022 onwards, the financial cycle contracts as inflationary pressures intensify and monetary policy tightens globally. The resulting increase in interest rates leads to a deterioration in financing conditions and a slowdown in credit dynamics.

4.2 Decomposition

Approach. The factor estimates can be further decomposed into contributions from the underlying variables, providing a clearer understanding of the main drivers of the financial cycle. The decomposition is performed in-sample as follows. We regress the estimated factor (obtained using the PCA estimator) on the standardised variables without a constant term. Each variable's contribution is then computed as the product of its normalised weight, defined as the absolute value of the corresponding regression coefficient divided by the sum of all absolute coefficients, and its observed value. An exception is made for sovereign bond spreads, whose weight retains the sign of the regression coefficient: because spread compression (declining spreads) signals financial cycle expansion, a negative coefficient on spreads correctly captures the inverse relationship between spread levels and the financial cycle, ensuring that this variable's contribution is correctly attributed.

Empirical evidence. Figure 3 illustrates the decomposition of the median financial cycle indicator across countries (results for individual countries are available in Appendix B). The financial cycle shows a pronounced build-up in the early 2000s, primarily driven by strong growth in household credit and credit to non-financial corporations, alongside rising DSR and credit-to-GDP ratios. This expansion is followed by a sharp contraction during the mid-2000s, as credit growth decelerates and the financial cycle indicator approaches its long-term average.

A significant increase occurs leading up to the 2007–2008 GFC, largely due to an increase in the growth of credit to households and increasing house prices. The onset of the GFC triggers a sharp and prolonged downturn in the financial cycle, a decline further exacerbated by the European sovereign debt crisis in 2010. During this period, the contribution of most variables, including credit indicators and house prices, turns negative, reflecting widespread deleveraging and financial distress across European economies.

In the second half of the 2010s, the financial cycle stabilises at relatively subdued levels, as credit growth remains weak despite gradual improvements. Notably, house prices emerge as a more significant driver of the financial cycle during this phase, reflecting growing overvaluation concerns in certain European housing markets. A renewed recovery is observed following the COVID-19 shock, driven by credit expansion and increased liquidity from unprecedented fiscal and monetary support.

Around 2022, the financial cycle enters a renewed downturn as global inflation accelerates and central banks implement rapid monetary tightening. The decline in the credit-to-GDP ratio and house prices weighs significantly on the financial cycle.

Overall, the decomposition underscores that while credit growth and leverage measures remain the primary drivers of financial cycle fluctuations, asset prices and interest rate spreads have played a significant role during specific periods. These insights can enhance the monitoring of financial cycles, help identify emerging trends and risks, and ultimately inform macroprudential policy.

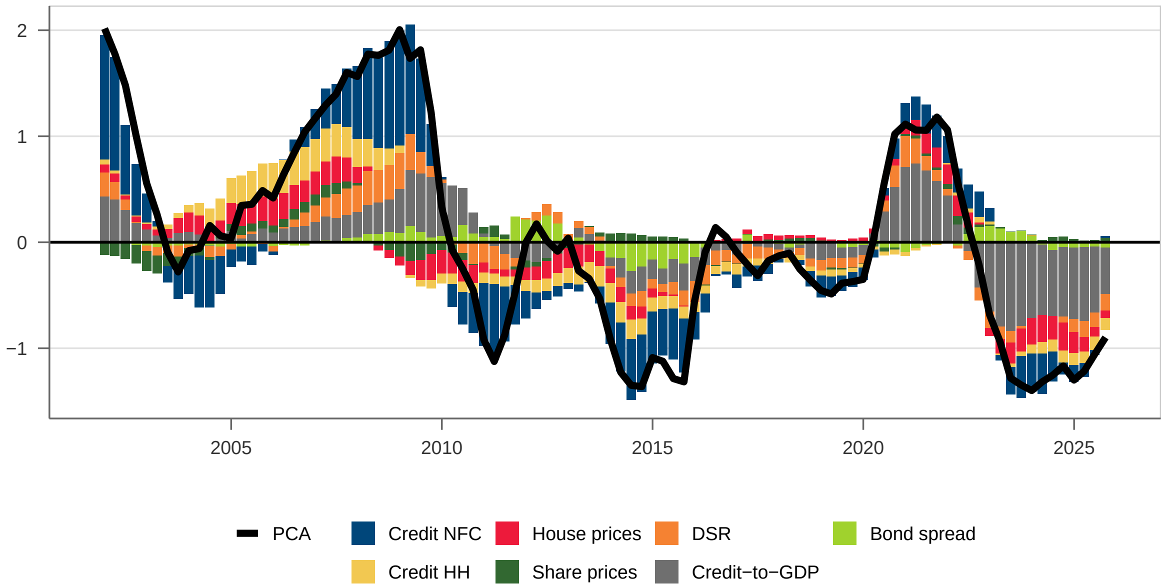

Figure 3: Decomposition of the median financial cycle indicator across countries

Notes: The figure shows the in-sample decomposition of the cross-country median financial cycle indicator. For each quarter, the median value of the PCA factor and each underlying variable is first computed across the 10 countries in the sample; the decomposition is then applied to this vector of median values. Each variable's contribution is the product of its observed (standardised) value and a normalised weight derived from a zero-intercept regression of the estimated factor on all variables. Weights are proportional to the absolute regression coefficients, except for sovereign bond spreads, which retain the sign of the coefficient to correctly reflect that spread compression signals financial cycle expansion.

4.3 Identification of a neutral risk environment

Approach. A neutral risk environment corresponds to a state in which cyclical systemic risk is neither elevated nor subdued. It is typically associated with moderate credit growth, stable asset price dynamics, and financial conditions that do not signal the build-up or materialisation of imbalances. In such periods, vulnerabilities evolve in line with underlying economic fundamentals, without clear signs of overheating or stress.

Since all underlying variables are standardised to have zero mean, periods where the financial cycle indicator is close to zero correspond to conditions in which the underlying variables are near their long-term averages. Our approach uses deviations from these averages to identify periods that are consistent with a neutral risk environment. Positive deviations from zero reflect conditions that are more buoyant than typical, suggesting a potential build-up of cyclical risk, while negative deviations indicate subdued conditions associated with the materialisation of risk or deleveraging. To assess when such deviations become statistically significant, we draw on the asymptotic distribution theory of the principal components estimator. This yields time-varying confidence bands around the financial cycle indicator: periods in which the indicator lies within these bands are classified as neutral, while statistically significant deviations above (below) the upper (lower) bound are classified as elevated (subdued).

Empirical evidence. Figure 4 illustrates this approach using the median financial cycle indicator across countries (country-specific results are reported in Appendix C). The shaded area represents the 95% confidence bands, while the dashed lines denote the time-averaged upper and lower confidence bounds, which serve as constant thresholds for regime classification. This framework allows for a clear classification of financial cycle phases. Elevated regimes correspond to periods in which the indicator exceeds the upper bound, signalling heightened systemic risk. Subdued regimes are identified when the indicator falls below the lower bound, typically reflecting the materialisation of risk and subsequent deleveraging. Neutral regimes correspond to periods in which the indicator remains within the confidence bands.

For the median across countries, the results identify elevated financial cycle phases in the early 2000s and in the run-up to the GFC, coinciding with strong credit expansion and rising asset prices. In contrast, the period following the European sovereign debt crisis is characterised by a subdued regime, reflecting weak credit growth and subdued financial activity. From around 2016 onwards, the financial cycle gradually converges towards a neutral regime. As of 2025 Q3, the most recent observations suggest that the financial cycle has moved back into a subdued phase.

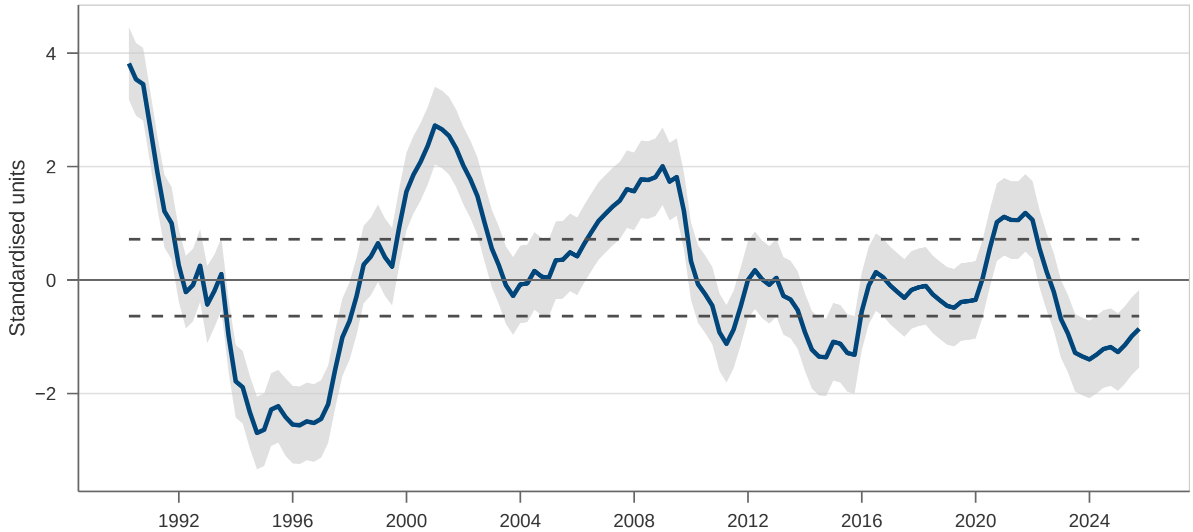

Figure 4: Financial cycle indicator and 95% confidence intervals

Notes: The figure displays the median financial cycle indicator across countries together with 95% confidence bands based on the asymptotic distribution of the estimated factor. The shaded area represents the time-varying confidence bands, while the dashed lines denote the time-averaged upper and lower bounds used as constant thresholds for regime classification. Values above (below) the upper (lower) dashed line indicate statistically significant deviations associated with elevated (subdued) cyclical systemic risk.

Cross-country analysis of financial cycle regimes. Extending this framework across countries, Figure 5 classifies financial cycle conditions into subdued, neutral, and elevated regimes over time. Subdued regimes are prevalent in the mid-to-late 1990s and during the post-GFC period, reflecting deleveraging and weak credit dynamics. Elevated regimes are concentrated in the early 2000s and in the run-up to the GFC, consistent with the build-up of systemic risk. Following the COVID-19 shock, many countries temporarily transition towards neutral conditions, supported by expansionary fiscal and monetary policies. Since 2022, however, several countries have returned to subdued regimes as tighter monetary policy and higher interest rates weigh on credit growth and financial activity.

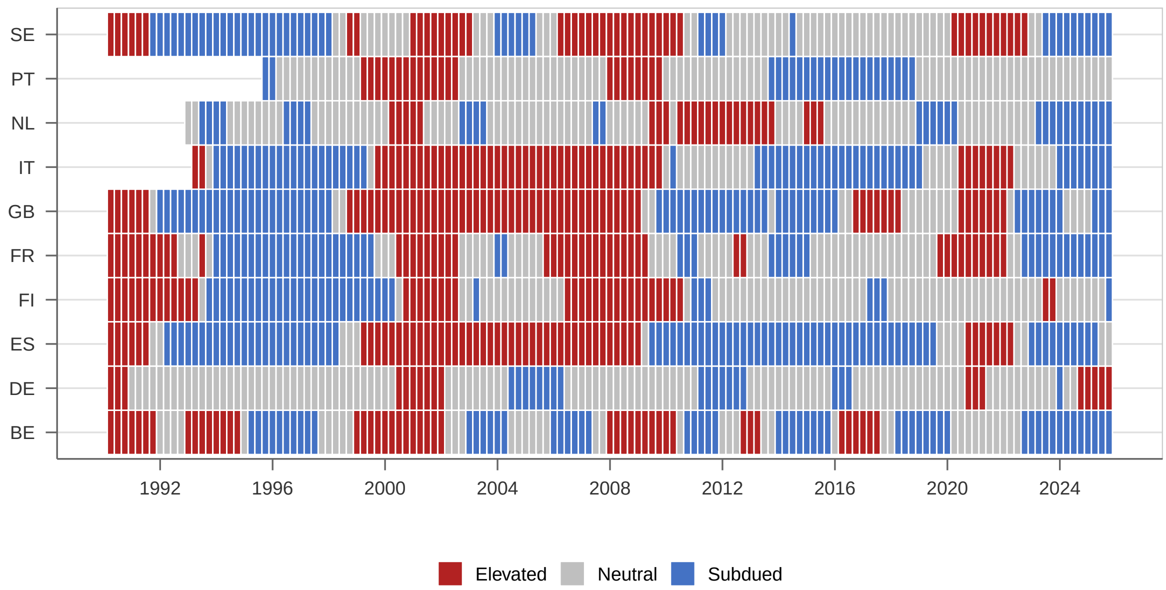

Figure 5: Financial cycle regimes across countries

Notes: The figure classifies financial cycle conditions across countries into subdued, neutral, and elevated regimes based on the position of the PCA financial cycle indicator relative to the time-averaged confidence bands for each country.

Policy implications. The calibration of the countercyclical capital buffer (CCyB) is subject to considerable uncertainty, as the build-up of cyclical systemic risk is difficult to measure in real time. In practice, this raises the risk of delayed or insufficient policy action, as well as the possibility of underestimating vulnerabilities during periods of apparent stability.

In response to these challenges, the concept of a positive neutral CCyB has gained prominence. As of November 2025, 17 macroprudential authorities in the European Economic Area (EEA) have implemented a positive neutral CCyB framework. A positive neutral CCyB refers to a capital buffer maintained when risks are assessed as neither elevated nor subdued, ensuring that banks hold releasable capital in normal times. Evidence from the COVID-19 episode shows that the release of such buffers supports lending and mitigates procyclical deleveraging, alleviating concerns about banks' reluctance to use capital buffers (Avezum et al., 2024; Berrospide et al., 2021; European Central Bank and European Systemic Risk Board, 2026).

Our approach enables the identification of periods in which cyclical systemic risks are neither building up nor materialising. By assessing deviations in the financial cycle, constructed as deviations of underlying indicators from their long-term averages, it provides a benchmark for evaluating when risks lie within a neutral range. In this context, the framework offers a transparent and operational tool to inform the calibration of the neutral level of the CCyB.

4.4 Identification of financial crises

As documented by Borio (2014), one defining feature of financial cycles is that they tend to be closely associated with systemic banking crises. This link suggests that a good measure of the financial cycle can be used as a leading indicator to signal the build-up of risk that may culminate in a financial crisis. In this section, we investigate the early-warning properties of our proposed measure of the financial cycle.

Our analysis is conducted using the systemic banking crisis dataset constructed by Lo Duca et al. (2017) for European countries. A crisis, as defined by this dataset, refers to a period during which the financial system served as a source or amplifier of economic shocks, financial intermediaries faced significant distress or collapse, and substantial crisis management policies were implemented.

Set-up of the early-warning analysis. We estimate country-specific logit regressions to assess the build-up of systemic risk ahead of banking crises. The dependent variable is a binary vulnerability indicator that takes the value one during the five to twelve quarters preceding a systemic banking crisis and zero otherwise. This definition emphasises the objective of detecting early signs of financial imbalances, rather than predicting the exact timing of a crisis. This distinction is important, as macroprudential policy must be preemptive to be effective. Moreover, there are implementation lags that constrain policy responsiveness. For example, the CCyB includes a one-year phase-in period, allowing banks time to adjust to new capital requirements. As a result, signals issued less than a year before a crisis are likely too late for an effective policy response. The focus is therefore on identifying vulnerabilities early enough to enable timely intervention.

As the independent variable, we consider the proposed financial cycle indicator estimated using three alternative estimators (PCA, QML, and two-step). Our analysis is purely univariate as we do not consider additional variables or lags of the indicator. To benchmark our results, we use the Basel gap, calculated as the cyclical component of the total credit-to-GDP ratio derived from a recursive HP filter with a smoothing parameter of 400,000. The Basel gap is chosen as the benchmark indicator because it is widely considered a leading univariate signal of systemic financial crises. It serves as the recommended reference indicator for activating the countercyclical capital buffer under the guidance of the Basel Committee on Banking Supervision.

We use two measures of accuracy in the early-warning exercise: the area under the receiver operating characteristic curve (AUROC); and the usefulness index proposed by Alessi and Detken (2011). The AUROC is a global measure of accuracy that relates the false positive rate (noise ratio) to the true positive rate (signal ratio) for every possible threshold value applied to the predicted probabilities from the logit regression. Hanley and McNeil (1982) show that the AUROC can be interpreted as the probability of correctly classifying zeros and ones.

The usefulness index represents a measure of the gain from following the model relative to disregarding it, adjusted for the preferences of policymakers regarding type I (failing to signal a financial crisis, denoted by T_1) and type II ("false alarms", denoted by T_2) errors. The index is defined as:

$$U = \frac{\min[\theta;\, 1-\theta] - L(\theta)}{\min[\theta;\, 1-\theta]}, \tag{16}$$

where $L(\theta)$ denotes the policymaker's loss function:

$$L(\theta) = \theta T_1 + (1-\theta)T_2. \tag{17}$$

The parameter θ controls policymaker preferences. A value of θ below 0.5 indicates that the policymaker is less averse to missing a crisis signal than to receiving a false alarm. Because the policymaker can always achieve a loss of $\min[\theta; 1-\theta]$ by disregarding the indicator, the usefulness index compares the loss under the logit model against the loss of ignoring it. We consider two preference specifications: balanced preferences ($\theta = 0.5$), following Lang et al. (2019), and crisis-averse preferences ($\theta = 0.7$), which assign greater weight to type I errors (failing to signal a crisis).

Finally, we consider the full sample used in the previous analysis, with two exceptions. First, Finland is excluded from the early-warning exercise, as it did not experience a systemic banking crisis during the sample period (1995 Q3–2025 Q3); its major banking crisis occurred in the early 1990s, prior to the start of the sample. Second, we exclude the crisis periods themselves from the sample used to estimate the logit models. This follows the recommendation of Bussiere and Fratzscher (2006), who show that early-warning model performance can be distorted by the disorderly and volatile adjustment of explanatory variables following the onset of a crisis. To avoid this so-called post-crisis bias, we focus only on the pre-crisis and tranquil periods in the estimation.

In-sample exercise. The results for all countries, presented in Table 3, show that the financial cycle indicator constructed using a dynamic factor model has good early-warning properties. On average, the DFM-based indicators outperform the Basel gap in terms of AUROC (0.68 for PCA versus 0.55 for BG), while delivering a comparable usefulness index under both balanced ($\theta = 0.5$) and crisis-averse ($\theta = 0.7$) preferences. The advantage is particularly marked for Germany, where the Basel gap delivers near-zero usefulness under crisis-averse preferences (0.02), while the PCA-based indicator remains informative (0.13). Among estimators, PCA and the two-step estimator perform similarly, while QML exhibits more variation across countries.

Table 3: In-sample early-warning exercise

	AUROC				**Usefulness** ($\theta = 0.5$)				**Usefulness** ($\theta = 0.7$)			
Country	**PCA**	**QML**	**Two-step**	**BG**	**PCA**	**QML**	**Two-step**	**BG**	**PCA**	**QML**	**Two-step**	**BG**
BE	0.51	0.65	0.57	0.51	0.20	0.20	0.21	0.26	0.12	0.12	0.13	0.15
DE	0.76	0.85	0.73	0.36	0.25	0.26	0.25	0.08	0.13	0.13	0.11	0.02
ES	1.00	0.93	0.99	0.60	0.49	0.46	0.48	0.28	0.29	0.27	0.29	0.17
FR	0.78	0.48	0.64	0.61	0.30	0.16	0.22	0.30	0.17	0.09	0.13	0.18
GB	0.72	0.77	0.77	0.63	0.29	0.29	0.30	0.27	0.18	0.18	0.18	0.16
IT	0.67	0.75	0.68	0.55	0.18	0.27	0.21	0.27	0.11	0.06	0.13	0.16
NL	0.42	0.62	0.45	0.53	0.07	0.19	0.10	0.27	0.04	0.11	0.06	0.16
PT	0.54	0.66	0.53	0.53	0.24	0.24	0.20	0.25	0.15	0.14	0.12	0.15
SE	0.73	0.57	0.67	0.64	0.26	0.10	0.18	0.32	0.14	0.06	0.11	0.19
Avg	**0.68**	**0.70**	**0.67**	**0.55**	**0.25**	**0.24**	**0.24**	**0.26**	**0.15**	**0.13**	**0.14**	**0.15**

Notes: PCA, QML, and Two-step refer to estimates of the financial cycle indicator obtained using the principal components, quasi-maximum likelihood, and two-step estimators, respectively. BG denotes the credit-to-GDP gap (Basel gap). AUROC is the area under the receiver operating characteristic curve. The usefulness index is computed following Alessi and Detken (2011); θ weights type I errors (failing to signal a crisis) against type II errors (false alarms). A value of $\theta = 0.5$ implies balanced preferences; $\theta = 0.7$ implies crisis-averse preferences (70% weight on missed crises). The maximum attainable usefulness is $\min(\theta, 1-\theta)$, equal to 0.50 for $\theta = 0.5$ and 0.30 for $\theta = 0.7$. Finland is excluded as it did not experience a systemic banking crisis during the sample period (1995 Q3–2025 Q3); its major banking crisis occurred in the early 1990s, prior to the start of the sample.

Simulated out-of-sample exercise. We also conduct an out-of-sample exercise focusing on the GFC. A key challenge for this evaluation is the very low frequency of systemic banking crises in the sample.

To address this issue, we follow an approach similar to Alessi and Detken (2018) and examine the model's predictions as of mid-2006, using only data available up to the second quarter of that year. Crucially, this exercise excludes any knowledge of the subsequent GFC, ensuring a realistic assessment of the model's practical utility. To broaden the set of countries for which out-of-sample predictions can be generated, we also include "residual events" as defined by Detken et al. (2014). Residual events are episodes in which domestic financial imbalances reached levels consistent with systemic stress, but an outright crisis was averted, typically because policy interventions or favourable external conditions dampened the credit cycle at a critical juncture. Treating these near-miss episodes as crisis signals allows us to draw on a richer set of early-warning observations.

The exercise is conducted in pseudo-real time, meaning that all transformations are based solely on information available up to each point in time. However, we acknowledge two simplifying assumptions: first, we rely on final revised data rather than vintages, and second, we assume no publication lag, meaning observations for a given period are assumed to be available immediately. The procedure involves estimating the dynamic factor model using data up to 2006 Q1. We then perform a logit regression of the vulnerability indicator on the financial cycle indicator and use the resulting model to predict the probability of a financial crisis occurring within one year. This process is repeated for each quarter through 2009 Q1.

Figure 6 illustrates the cross-country median of the predicted crisis probabilities alongside the interquartile range (25th–75th percentiles). The key finding is the trajectory of the indicator: the estimated median probability of a systemic banking crisis within one year approximately doubles between early 2006 and 2007, rising from approximately 10% to around 20%. This sharp increase, generated entirely from information available in real time and without any knowledge of the forthcoming GFC, constitutes a clear and economically meaningful early-warning signal. That the absolute probability level does not exceed 20% is unsurprising: the logit models are estimated on a sample that had not yet witnessed a crisis of the GFC's magnitude, so the predicted probabilities are naturally anchored to the historical base rate of crisis occurrence. What matters for policy purposes is the signal embedded in the trend. An approximate doubling of the estimated crisis probability in the two years preceding the crisis represents a meaning-

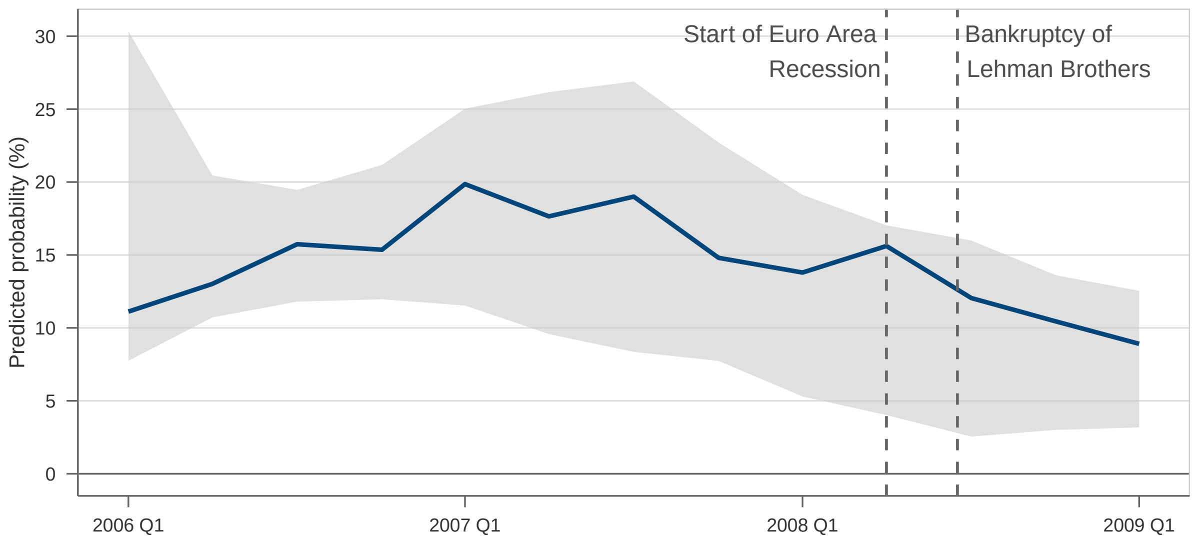

Figure 6: Simulated out-of-sample predicted probability of a financial crisis within one year

Notes: The figure reports the cross-country median predicted probability of a systemic banking crisis within one year, based on the PCA financial cycle indicator. The shaded area represents the interquartile range (25th–75th percentiles) of predicted probabilities across countries at each point in time. Predictions are obtained from country-specific logit models estimated in pseudo-real time using only information available up to each point in time. The left dashed vertical line marks the start of the euro area recession (2008 Q2); the right dashed vertical line marks the bankruptcy of Lehman Brothers (15 September 2008).

ful early-warning signal for macroprudential policy. These results suggest that even a simple univariate model based on the financial cycle indicator can provide valuable and actionable early-warning signals.

5 Conclusion

This paper proposes the use of dynamic factor models to construct country-specific financial cycle indicators, capturing the common dynamics of variables related to credit, asset prices, indebtedness, and financial conditions. The proposed indicator explains a substantial share of the variance in these variables, effectively representing key aspects of financial cycle fluctuations.

Our findings show that the indicator provides a coherent narrative of the financial cycle's evolution across Europe, aligning closely with historical episodes of financial expansion and contraction. Moreover, it demonstrates strong early-warning capabilities, particularly in signalling heightened systemic risk in the lead-up to the GFC. In a simulated out-of-sample exercise, the estimated probability of a systemic banking crisis within one year more than doubles in the period immediately before the GFC, rising from approximately 10% in early 2006 to around 20% by 2007.

We also introduce a methodology to decompose the indicator into contributions from individual variables and define data-driven thresholds for identifying periods of neutral systemic risk. These tools offer valuable insights for monitoring cyclical systemic risk and informing macroprudential policy, particularly for the calibration of the countercyclical capital buffer.

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A Factor model estimates

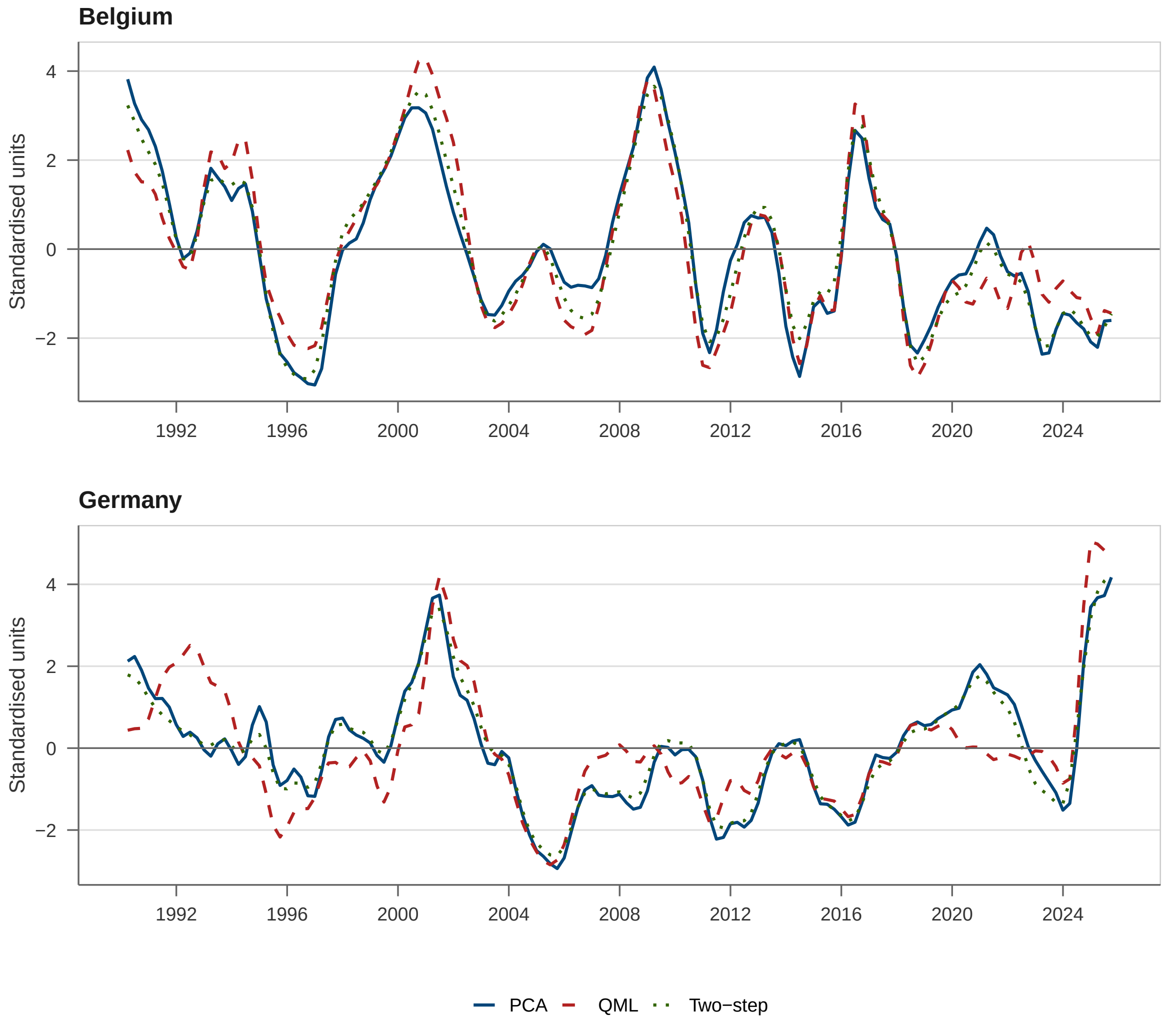

Figure 7: Financial cycle estimates: Belgium and Germany

Notes: The figure reports the estimated financial cycle for Belgium (top) and Germany (bottom), obtained using three estimation methods: principal components (PCA), quasi-maximum likelihood (QML), and the two-step estimator. The horizontal dashed line indicates zero.

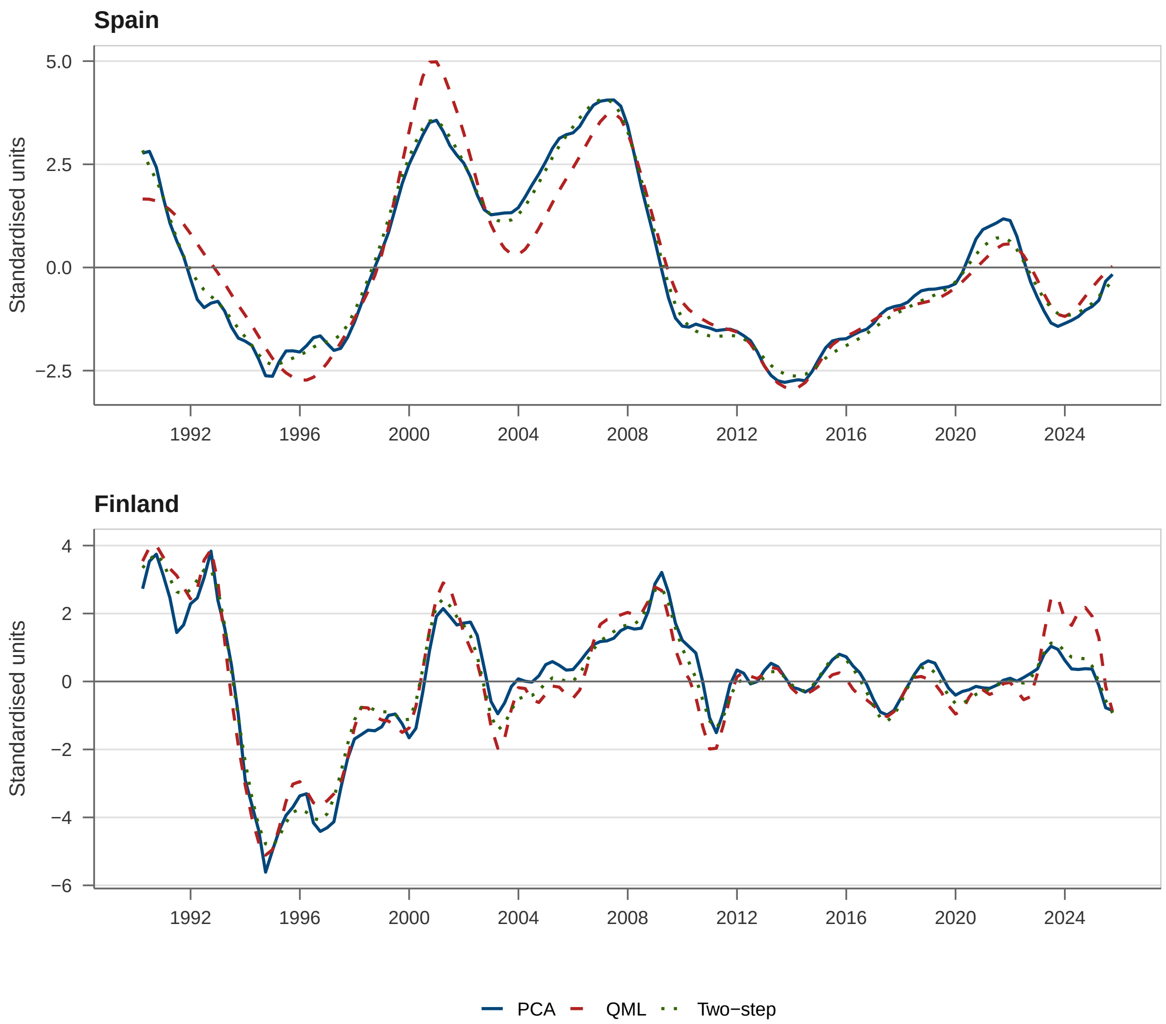

Figure 8: Financial cycle estimates: Spain and Finland

Notes: See notes to Figure 7. Top panel: Spain (ES). Bottom panel: Finland (FI).

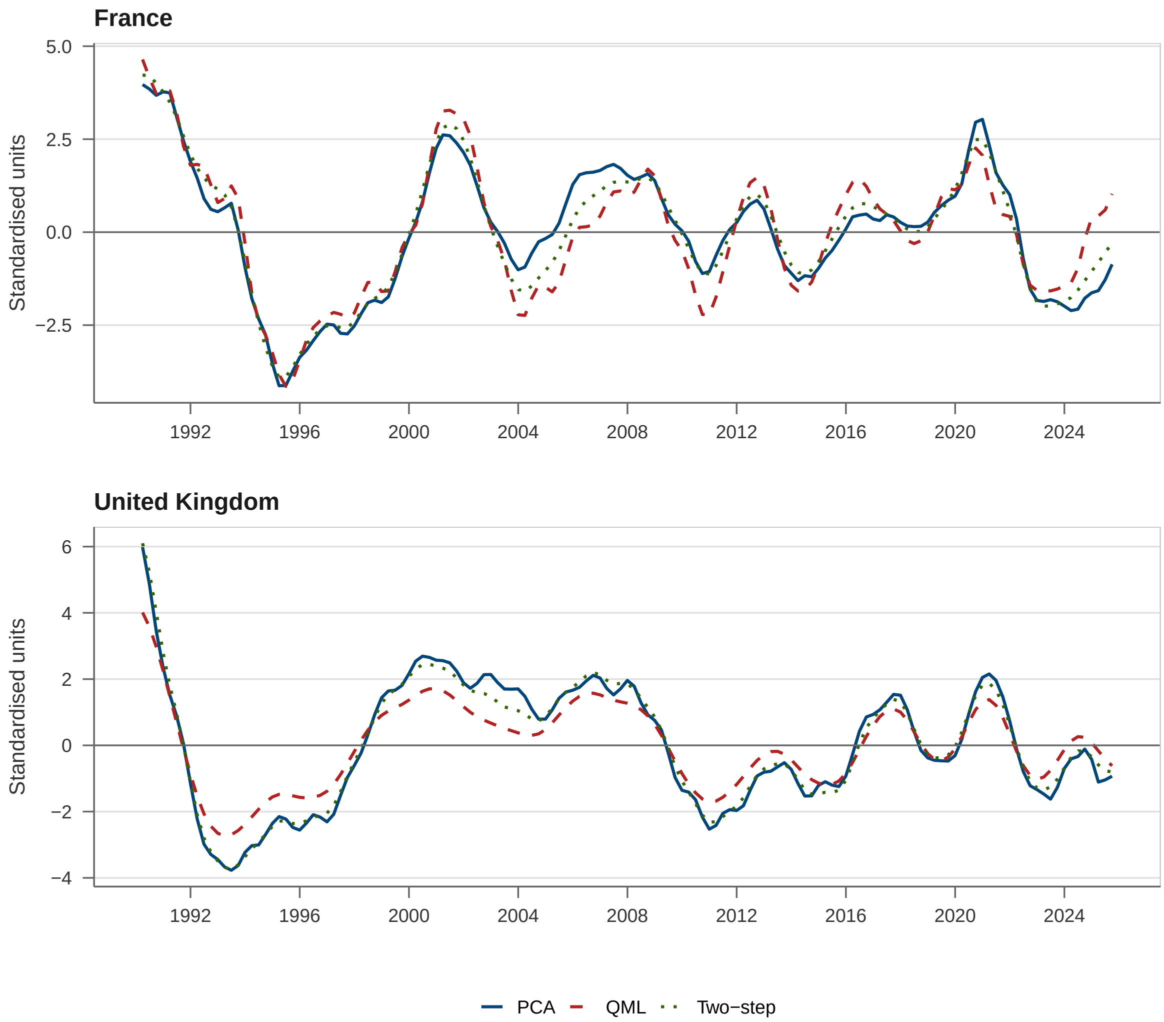

Figure 9: Financial cycle estimates: France and United Kingdom

Notes: See notes to Figure 7. Top panel: France (FR). Bottom panel: United Kingdom (GB).

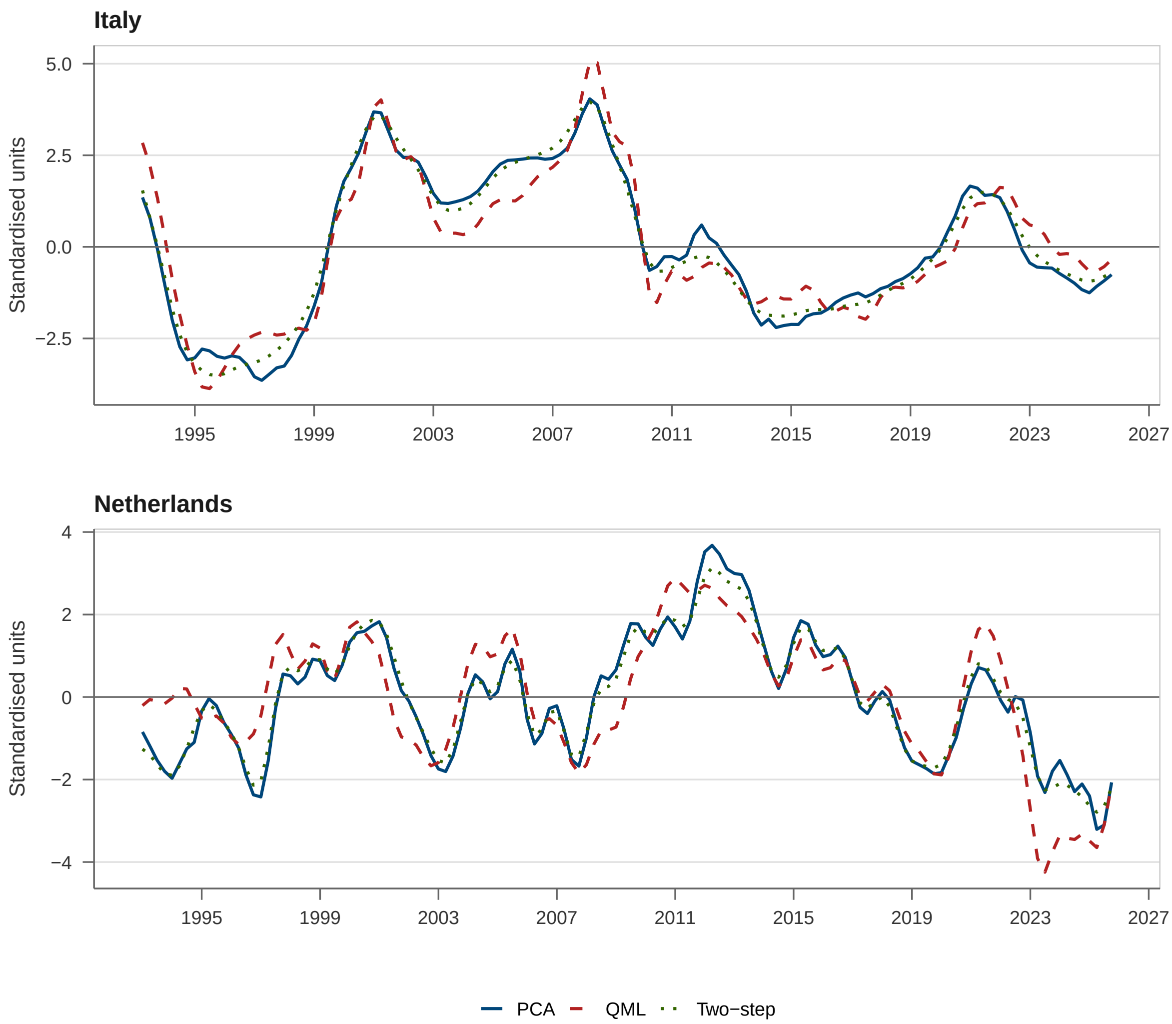

Figure 10: Financial cycle estimates: Italy and Netherlands

Notes: See notes to Figure 7. Top panel: Italy (IT). Bottom panel: Netherlands (NL).

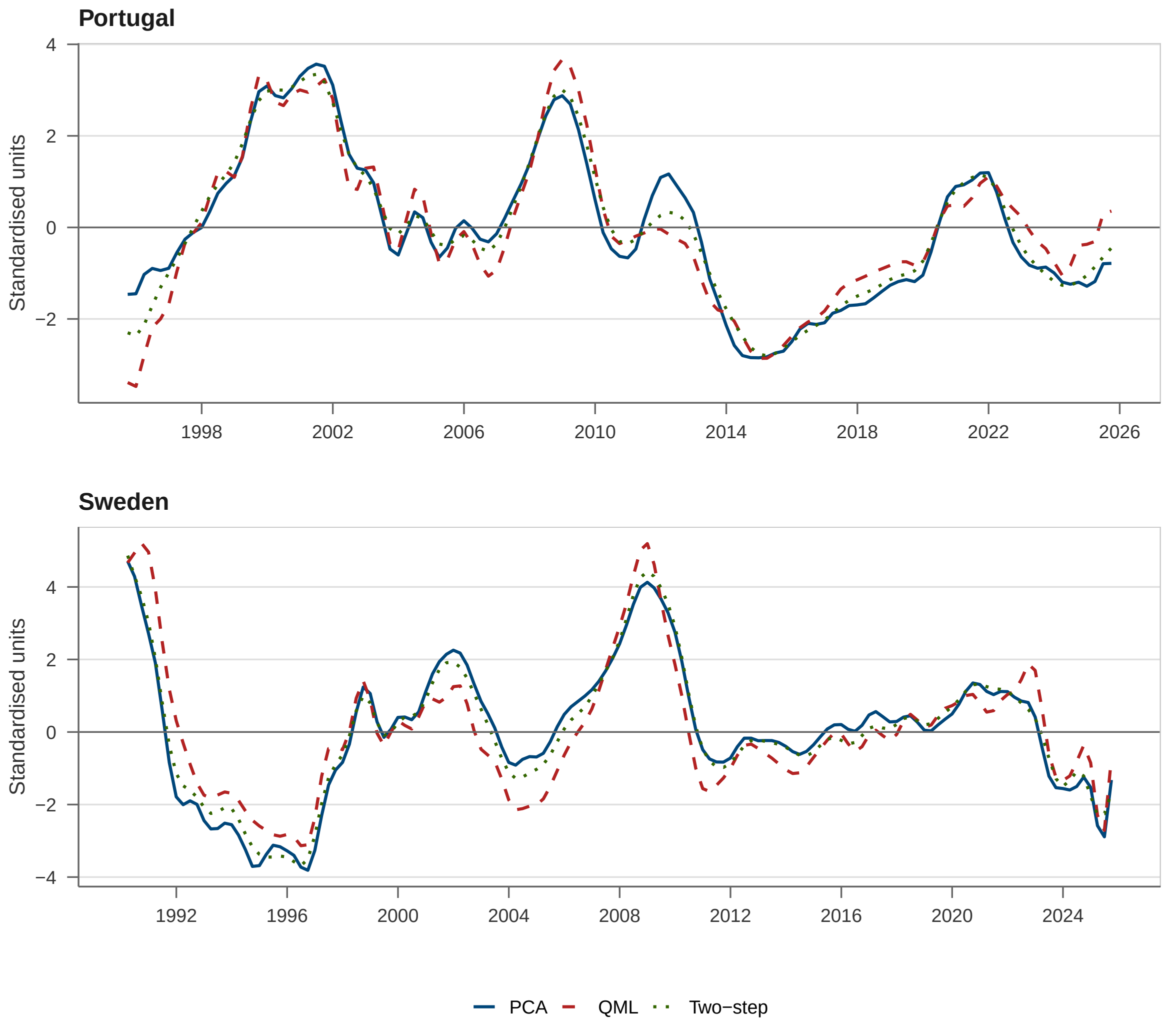

Figure 11: Financial cycle estimates: Portugal and Sweden

Notes: See notes to Figure 7. Top panel: Portugal (PT). Bottom panel: Sweden (SE).

B Financial cycle decomposition

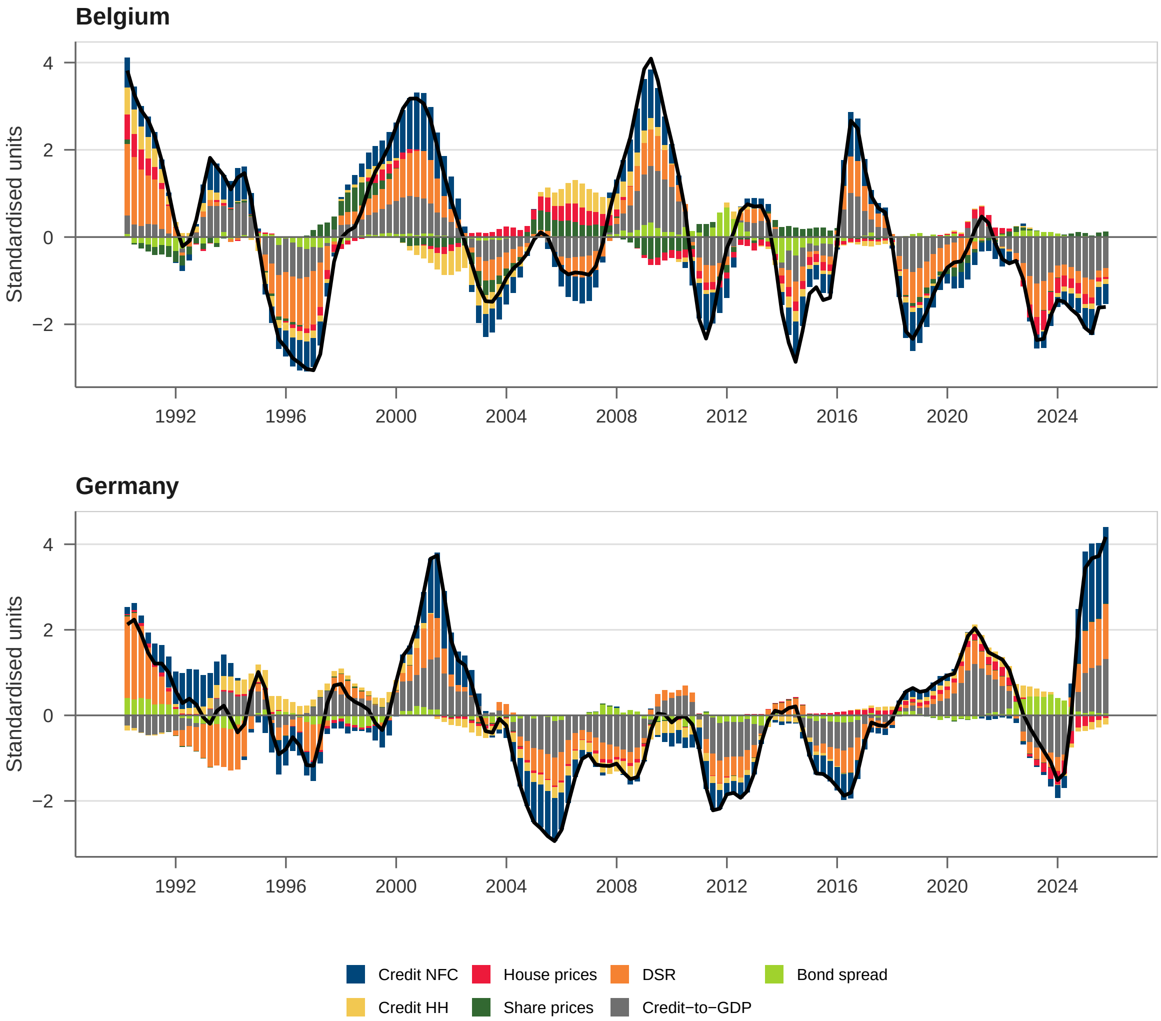

Figure 12: Financial cycle decomposition: Belgium and Germany

Notes: The figure shows the in-sample decomposition of the financial cycle indicator into contributions from the underlying variables for Belgium (top) and Germany (bottom). Bars represent weighted variable contributions; the black line is the PCA factor estimate. See the notes to Figure 3 for details on the decomposition methodology.

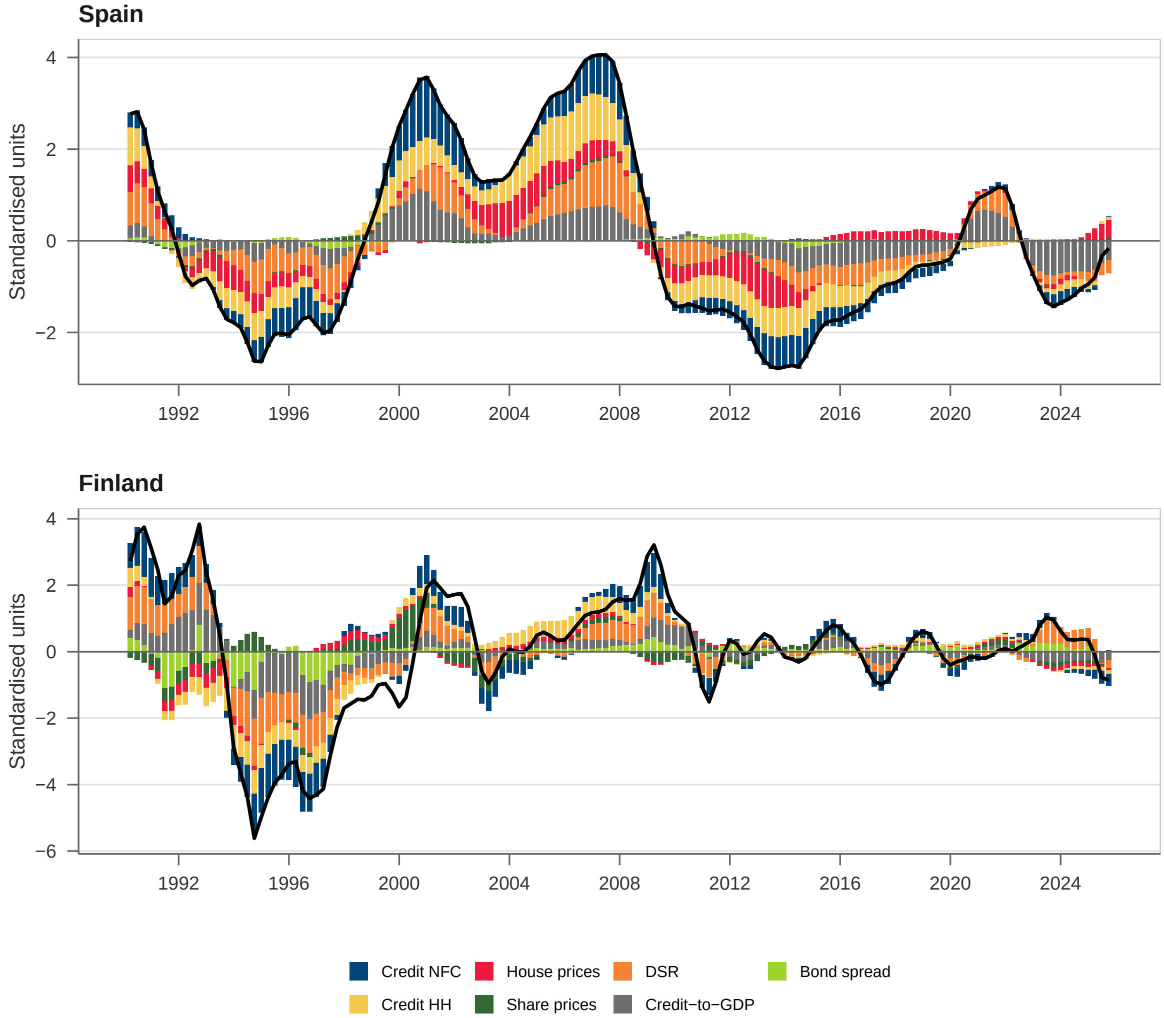

Figure 13: Financial cycle decomposition: Spain and Finland

Notes: See notes to Figure 12. Top panel: Spain (ES). Bottom panel: Finland (FI).

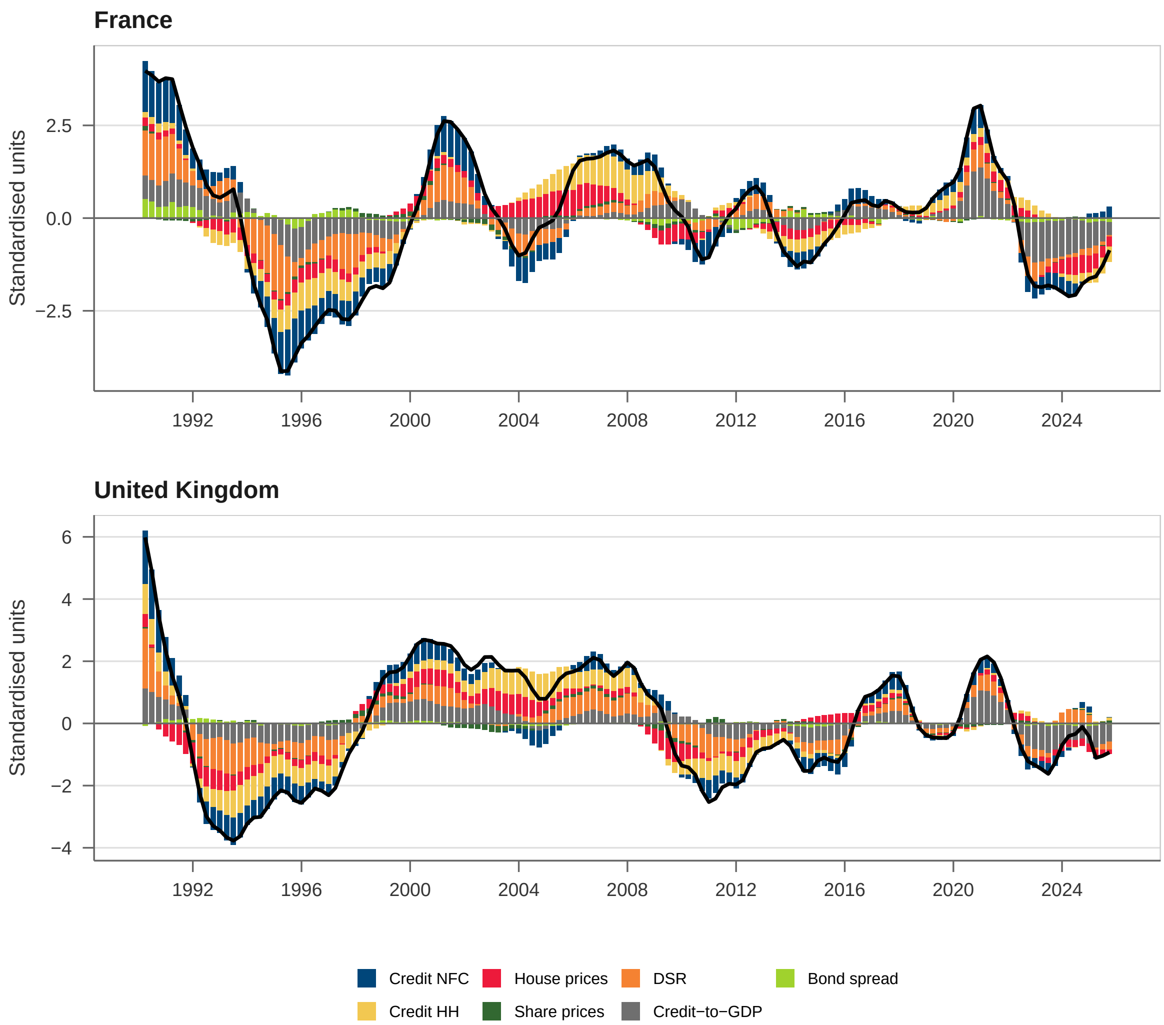

Figure 14: Financial cycle decomposition: France and United Kingdom

Notes: See notes to Figure 12. Top panel: France (FR). Bottom panel: United Kingdom (GB).

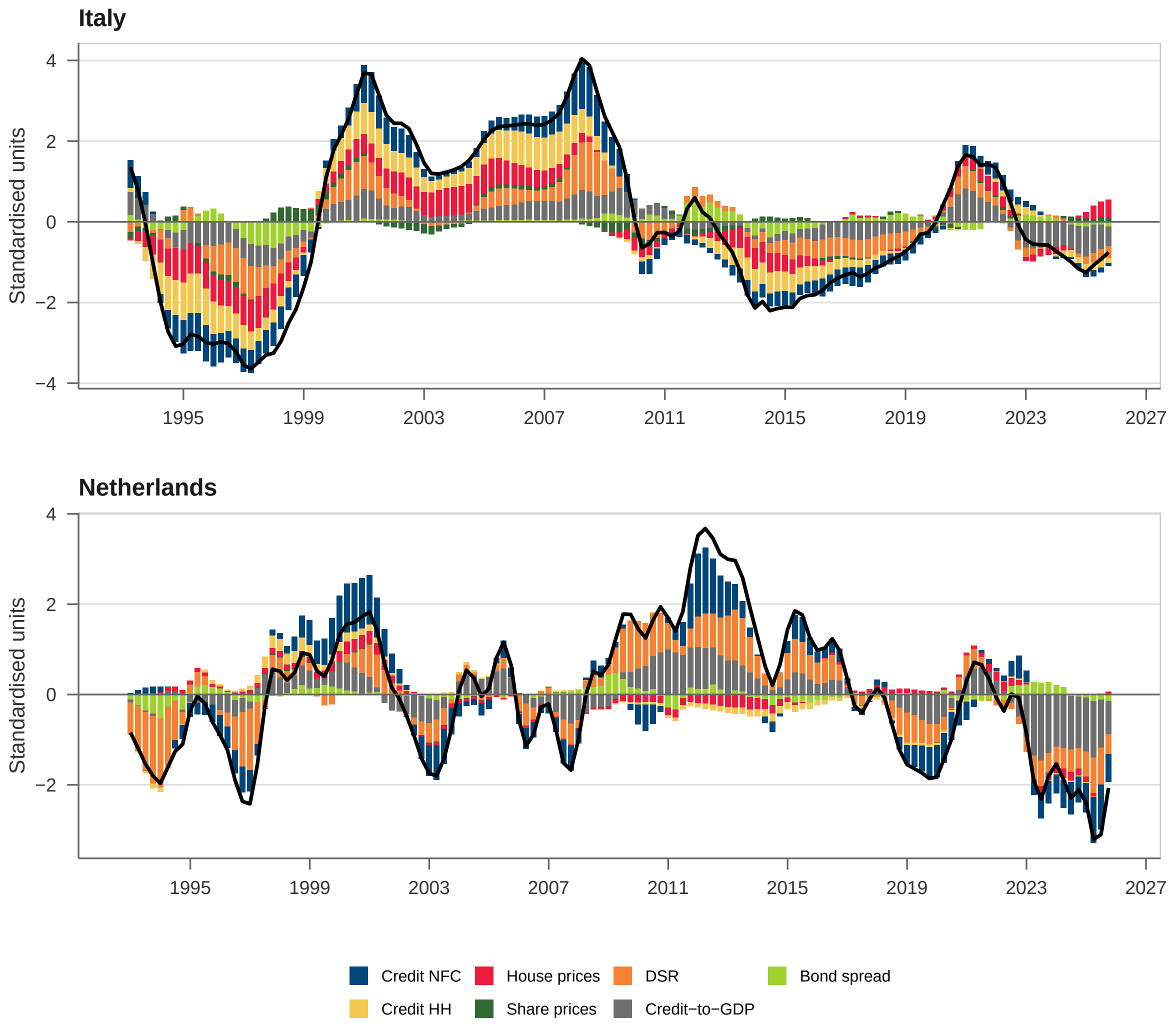

Figure 15: Financial cycle decomposition: Italy and Netherlands

Notes: See notes to Figure 12. Top panel: Italy (IT). Bottom panel: Netherlands (NL).

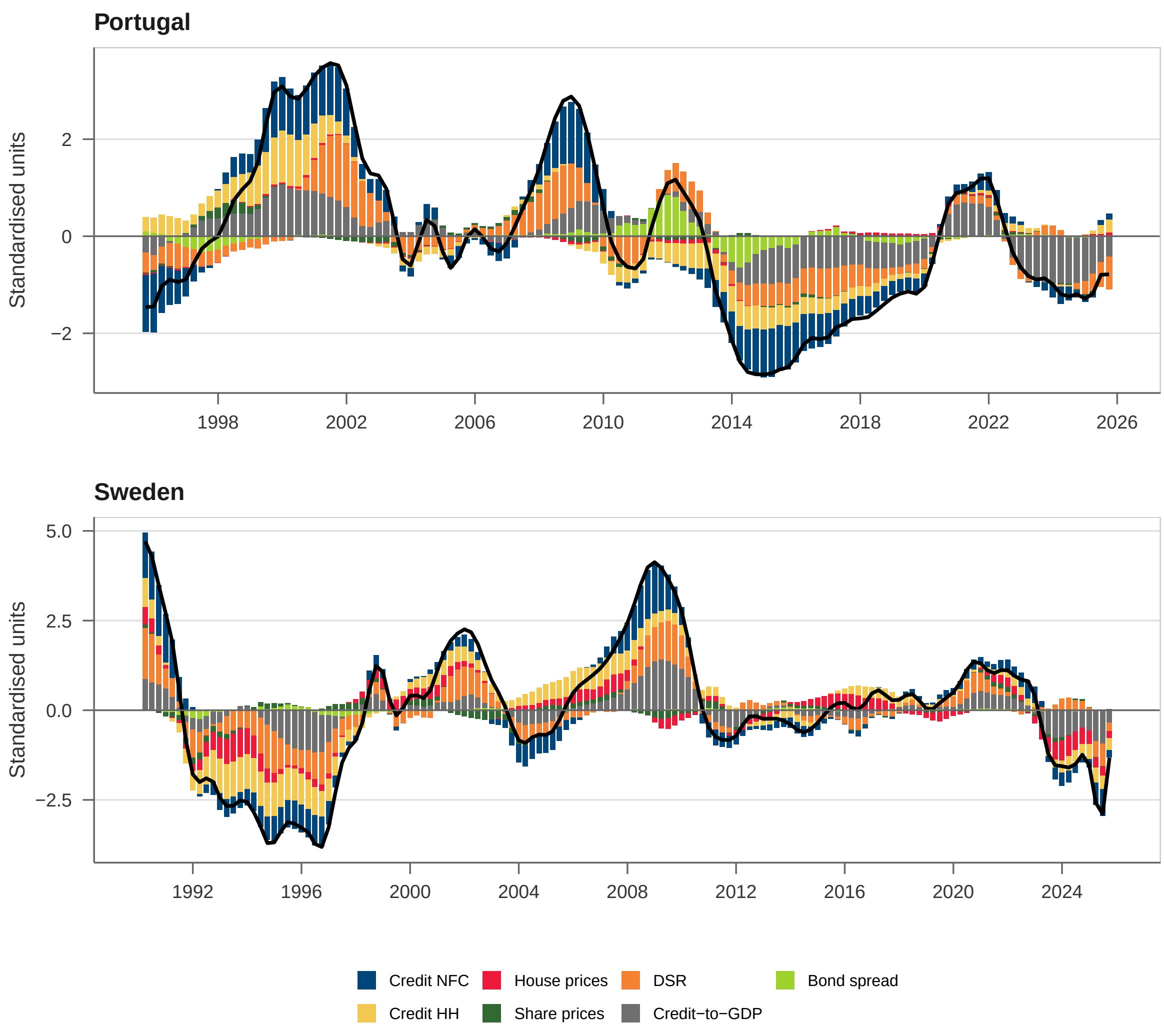

Figure 16: Financial cycle decomposition: Portugal and Sweden

Notes: See notes to Figure 12. Top panel: Portugal (PT). Bottom panel: Sweden (SE).

C Factor model confidence intervals

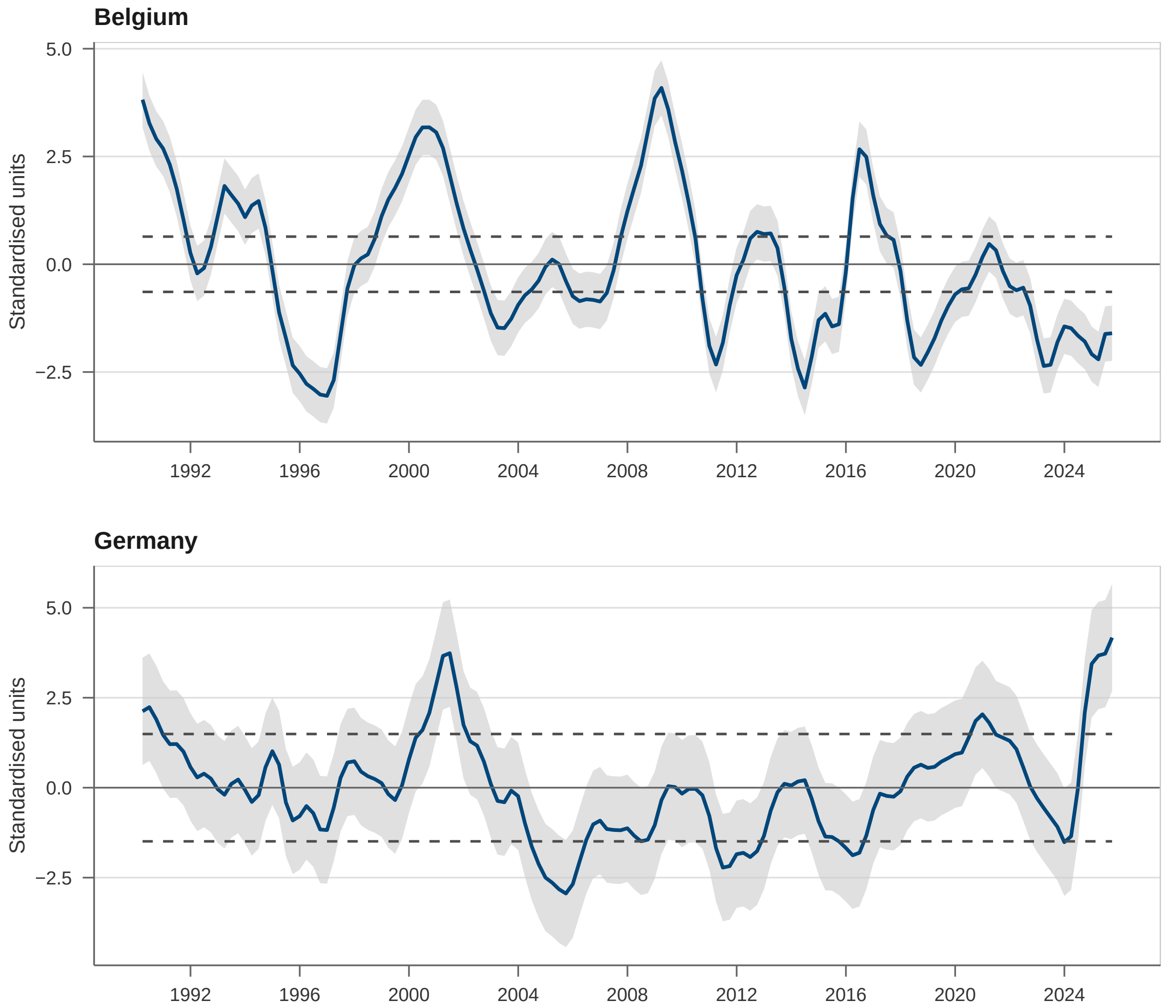

Figure 17: Financial cycle and confidence intervals: Belgium and Germany

Notes: The figure displays the estimated financial cycle (solid line) for Belgium (top) and Germany (bottom), together with 95% confidence bands (shaded area) based on the asymptotic distribution of the PCA estimator. The dashed lines indicate the mean of the upper and lower confidence bounds, which are used to classify financial cycle regimes. Values above (below) the upper (lower) dashed line indicate statistically significant deviations associated with elevated (subdued) cyclical systemic risk.

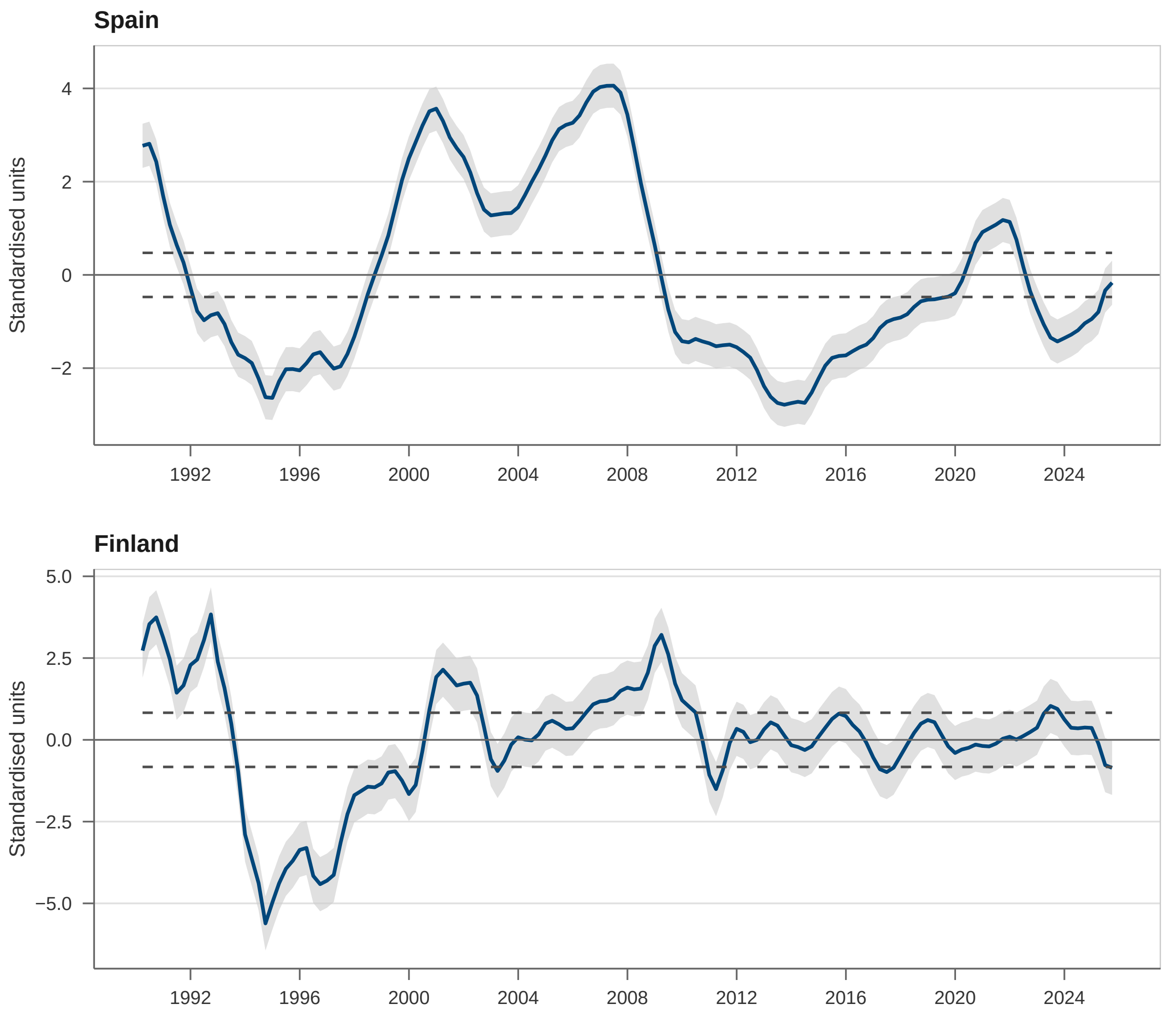

Figure 18: Financial cycle and confidence intervals: Spain and Finland

Notes: See notes to Figure 17. Top panel: Spain (ES). Bottom panel: Finland (FI).

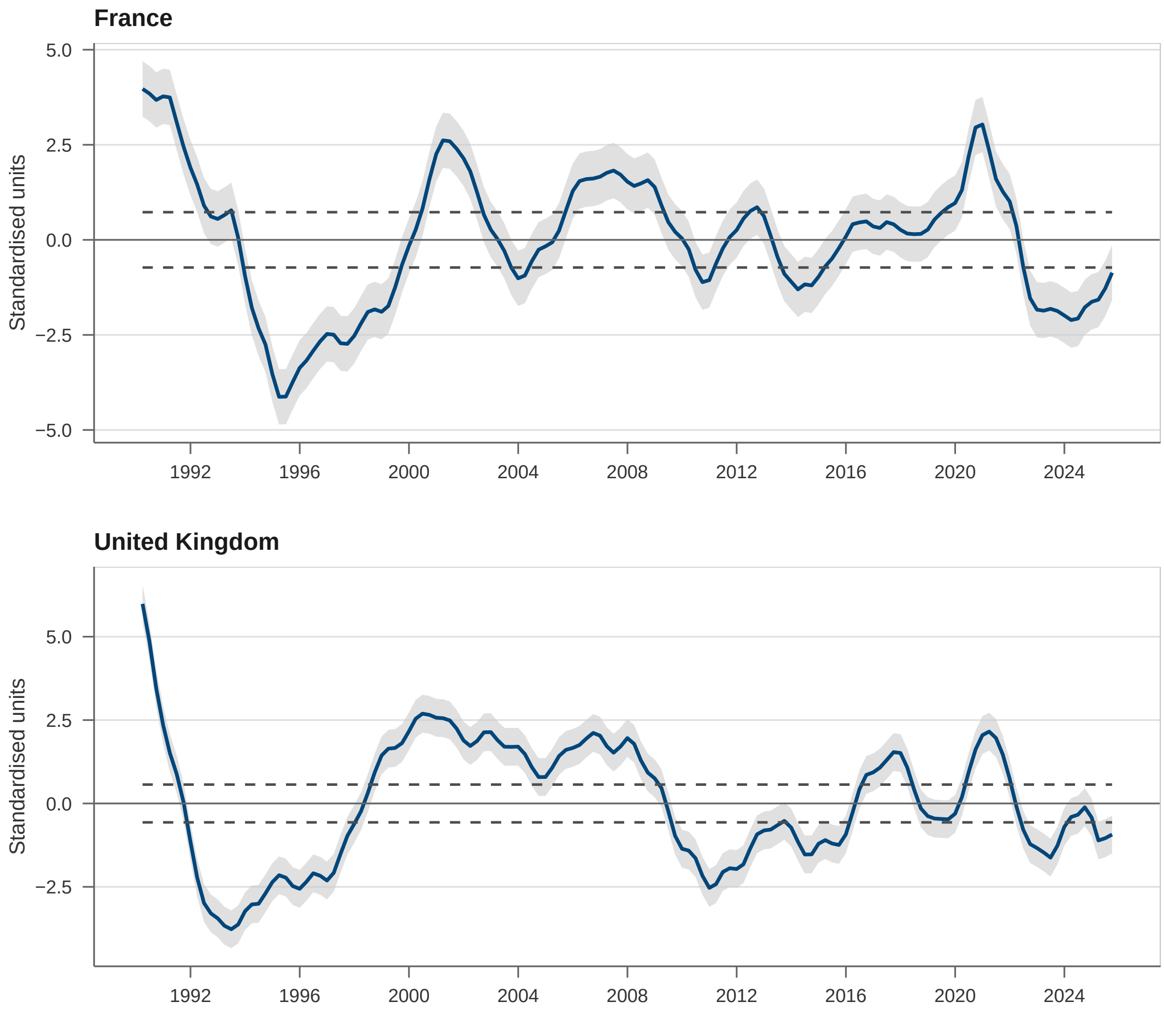

Figure 19: Financial cycle and confidence intervals: France and United Kingdom

Notes: See notes to Figure 17. Top panel: France (FR). Bottom panel: United Kingdom (GB).

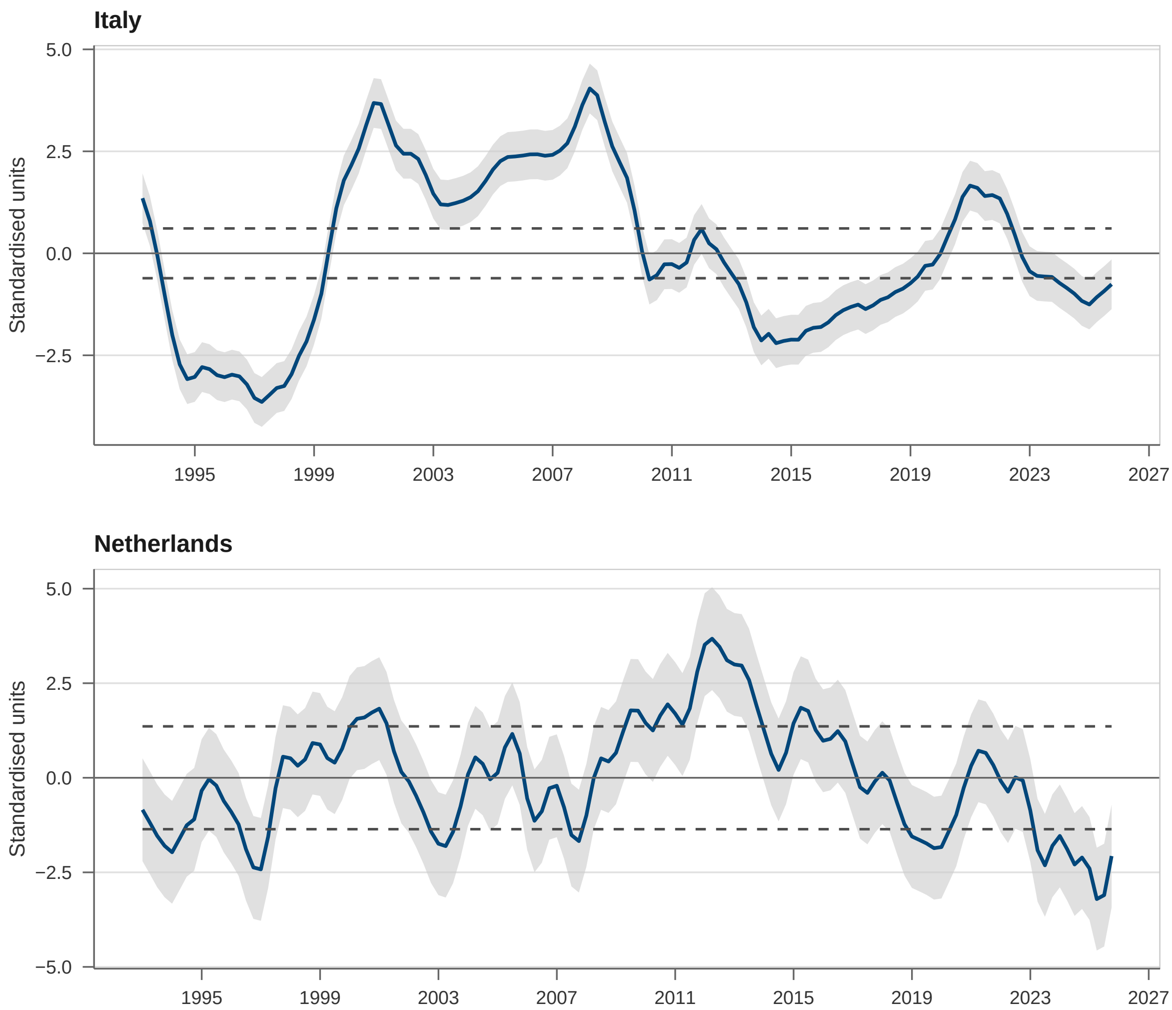

Figure 20: Financial cycle and confidence intervals: Italy and Netherlands

Notes: See notes to Figure 17. Top panel: Italy (IT). Bottom panel: Netherlands (NL).

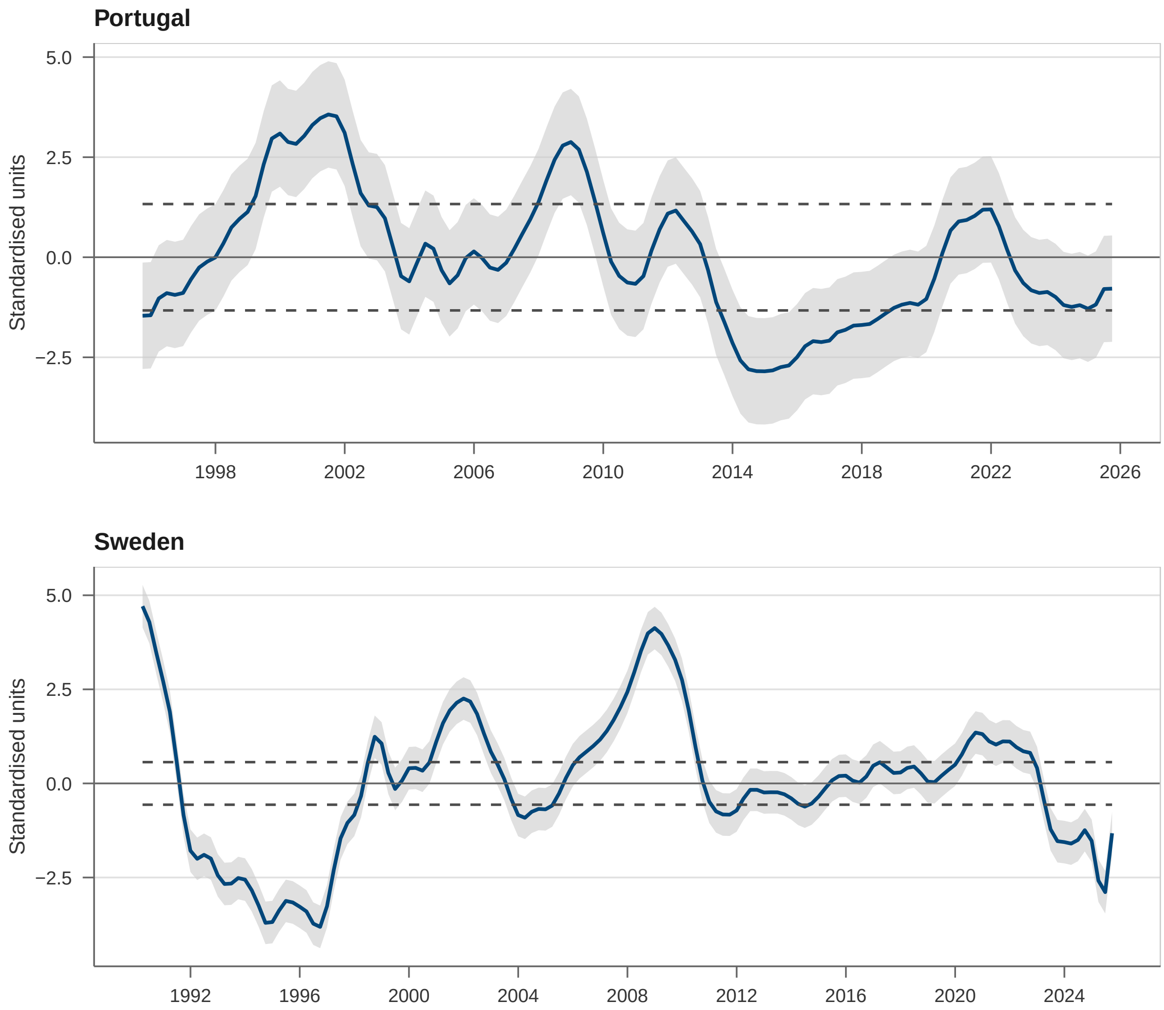

Figure 21: Financial cycle and confidence intervals: Portugal and Sweden

Notes: See notes to Figure 17. Top panel: Portugal (PT). Bottom panel: Sweden (SE).